

Module 1

Organization

Organization is a pattern of relationships among the individuals working together for a common goal.

The systems approach considers organization as a system composed of sub systems that are inter-related. Systems have boundaries, but they also interact with external environment. That is, they are open systems. This approach recognizes the importance of studying inter-relatedness of planning, organizing, and controlling in an organization as well as the many subsystems.

Characteristics of Organization

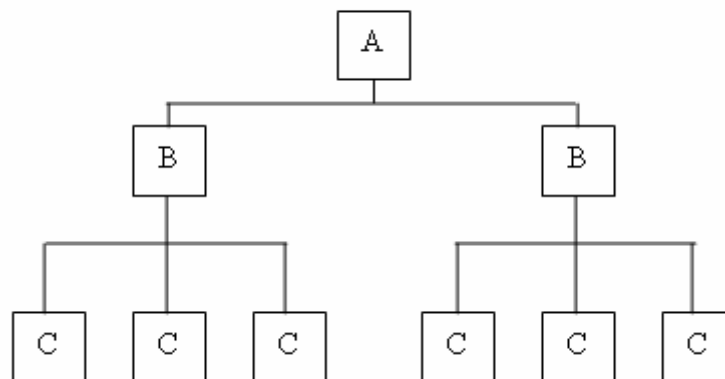
- 1) Organization is made up of a group of people
- 2) The group works under an executive head
- 3) Organization is a tool of management
- 4) It leads to division of work and responsibilities
- 5) It defines and fixes the duties and responsibilities of employees
- 6) It establishes a relationship between authority and responsibility and controls the effort of the group
- 7) Organization is a step towards achievement of established goals

Elements of Organization

The main elements or components of an organization are:

- 1) Well defined objectives
- 2) Well organized and coordinated group of people
- 3) Proper division of work and labour
- 4) Clear and well defined policies and procedures
- 5) Proper division of authority and responsibility
- 6) An effective system of communication

Organization Chart



Simple Organization Chart

An organization chart is a simple drawing of lines and boxes showing how the firm is organized. Boxes represent the activities of the firm and the people who perform these activities. Lines indicate the relationships among them. Positions near the top of the chart have more authority and responsibility than those below them. The number of horizontal rows of boxes will indicate the levels of management in an organization. The above figure shows a sample organization chart.

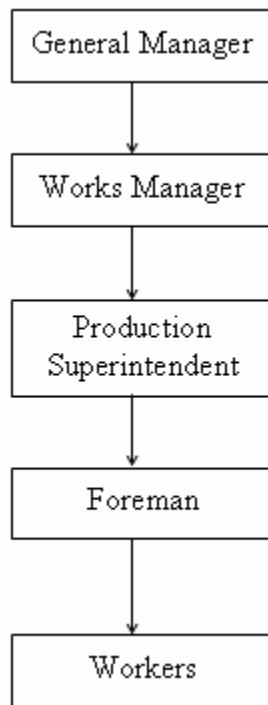
Types of Organization

A few commonly known forms of organization structures or types of organization are:

- 1) Line organization
- 2) Line and staff organization
- 3) Functional Organization
- 4) Project Organization, and
- 5) Matrix Organization

Line Organization

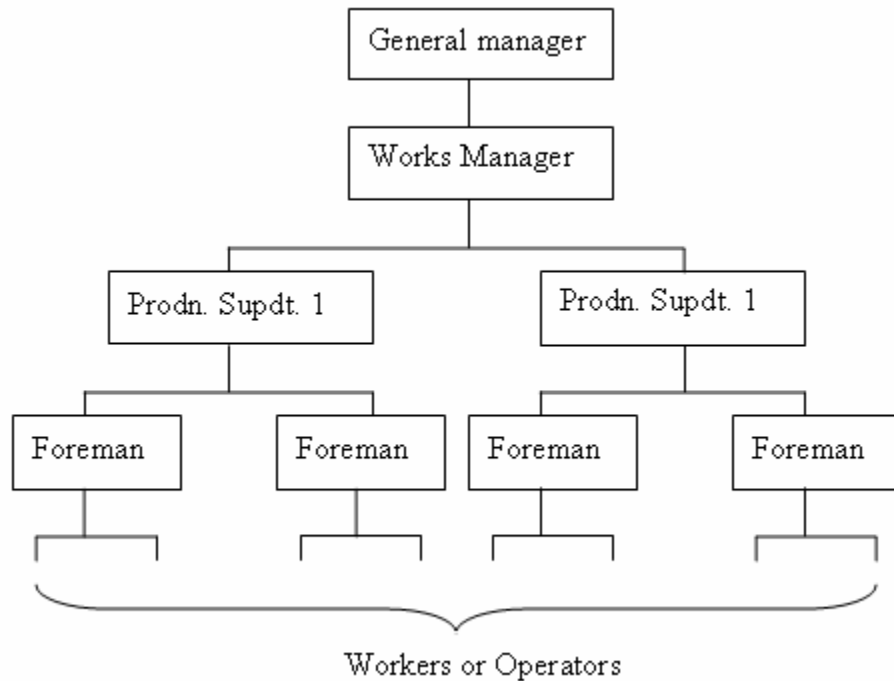
Line structure is historically the oldest type, and all other kinds of structures are modifications of line structures. This structure is characterized by the direct vertical flow of authority from top to bottom. A simplified line structure is shown in the following figure.



Line Structure

In this structure, the authority flows from the General Manager to Works Manager to production Superintendent to Foreman and to Workers.

Line Organization is also called military or scalar organization. Line organization is suitable for small concerns and for automatic and continuous process industries such as paper, sugar, cement, textile, etc.



Line structure

Advantages of Line structure

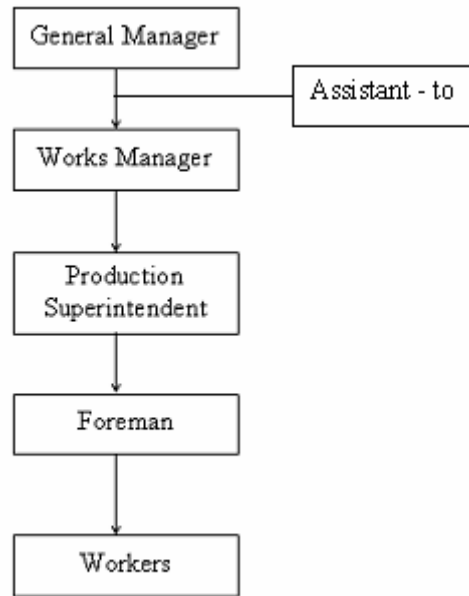
1. *Simplicity and clarity*
2. *Clear cut authority and responsibility*
3. *Strong discipline*
4. *Capable of developing all round executives at higher levels of authority*

Disadvantages of Line structure

1. *Neglects specialists*
2. *Lack of specialization may lead to wastage of materials as well as man and machine hours*
3. *Overloads a few important executives*
4. *Encourages dictatorial way of working*
5. *Limited to very small concerns*

Line and Staff Organization

Line and staff organization is a development of line organization. In this type of structure, special executives known as staff are employed to assist the line executives. Staff personnel act as helpers to the line and, as such, have no direct authority. The nature of staff relationship is advisory.



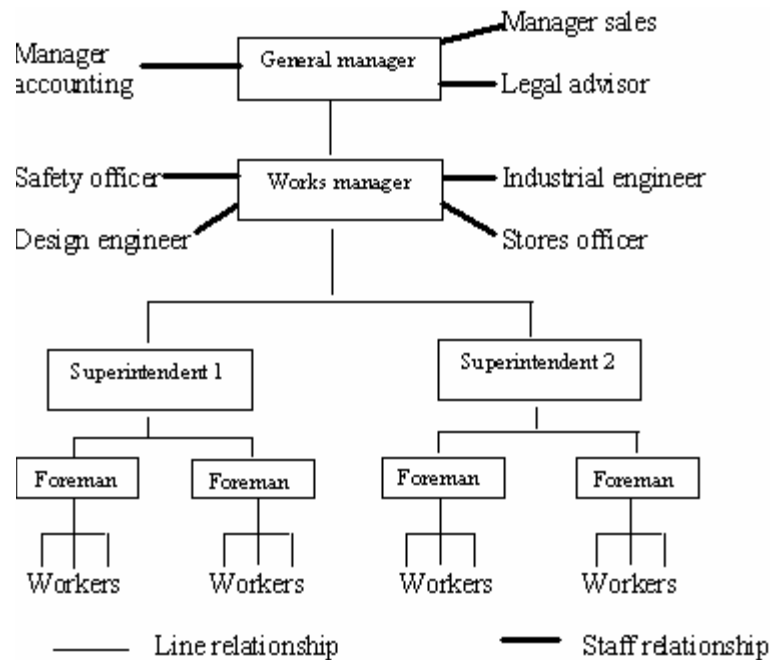
Line and Staff Organization

Advantages

1. Specialization benefits of staff can be profitably utilized to have standard operations.
2. Line executives are relieved of some of their workloads and are thus able to concentrate on other important matters.
3. Less wastage of material and labour
4. Improved product quality
5. Relatively flexible

Disadvantages

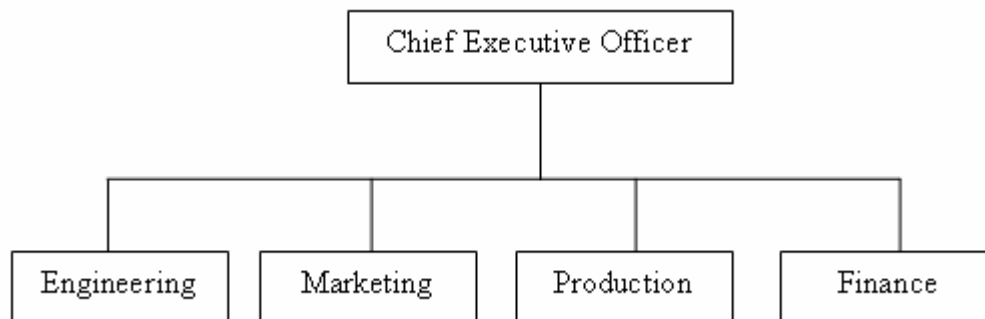
1. Staff – line conflict
2. Paper work may be increased very much
3. Staff men may dominate over the lower the lower-level line managers
4. Increased product cost because of high salaries of staff executives
5. Too much staff activity may complicate a line executive's job of leadership and control



Line and Staff Organization

Functional Structure

In a functional structure activities are grouped in accordance with the functions of an enterprise. This specialization leads to greater efficiency and refinement of particular expertise. The functional structure helps to focus on those departments that are critical for the success of the enterprise. The following figure shows the model of a functional structure.



Functional Structure

Advantages

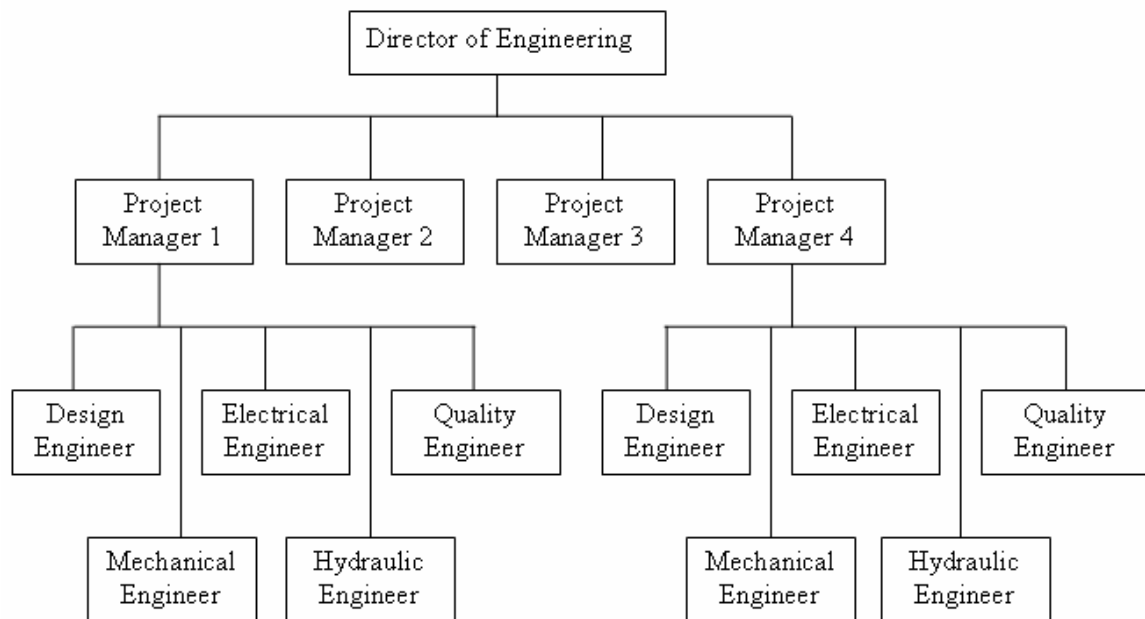
1. *Efficient use of resources*
2. *Simplifies training*
3. *Promotes professional development*
4. *Centralized control of strategic decisions*
5. *Improved quality of work*

Disadvantages

1. *Limits development of general managers or all-round executives*
2. *Restricted view of company objectives*
3. *Difficulty in multifunctional decision making*
4. *Promotes narrow specialization*
5. *Makes industrial relationships more complex*

Project Organization

Firms dealing with multiple products or different projects usually adopt project organization. In this type of organization, a project manager is put in charge of all engineering and support personnel necessary to accomplish an entire project. The emphasis in project organization is on creation of teams for the accomplishment of specific objectives. The following figure shows a project organization in Engineering.



Project Organization

Advantages

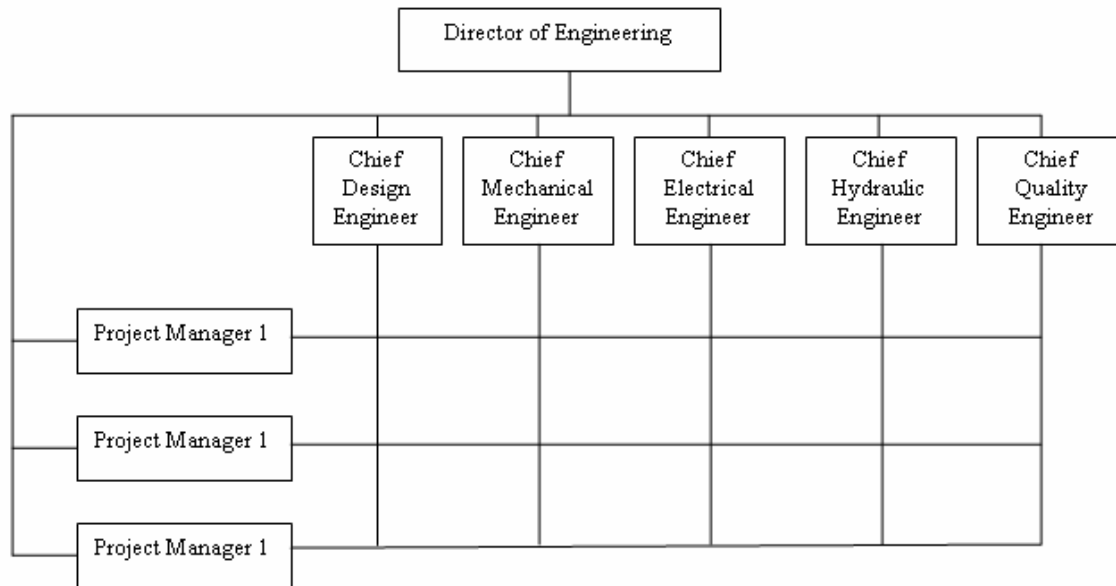
1. *Flexibility*
2. *Responsive to changing environment*
3. *Encourages team work*

Disadvantages

1. *Projects can be of short duration leading to frequent change in organization structure*
2. *Professionals prefer to be allied with their professional group rather than being allied with a project*
3. *Trained professionals need not tolerate the insecurity of frequent organization change*

Matrix Organization

The matrix organization is an organization structure that establishes two chains of command, one vertical and one horizontal, at the same time. It is intended to combine the advantages of functional structure and project structure. The following figure shows a matrix organization in Engineering.



Matrix Organization

In the matrix organization shown above the chiefs of design, mechanical, electrical, hydraulic and quality represent the functional departments that make up the vertical hierarchy. Simultaneously, the managers of projects 1,2 and 3 operate across the structure. This graphically creates a grid or matrix.

Advantages

1. *Decentralized decision making*
2. *Efficient use of functional managers*
3. *Capable of adapting to fast environmental changes*
4. *Flexibility*

Disadvantages

1. *Violates the principle of unity of command*
2. *High administrative costs*
3. *Requires tremendous horizontal and vertical co-ordination*
4. *Chances of interpersonal conflicts*

Authority and Responsibility

Authority is the organization's legitimized power that is linked to each position within the organization. It typically involves the right to command, to perform, to make decisions, and to expend resources.

Four management theories have attempted to explain the nature of authority. The classical theory of authority holds that authority flows from top to bottom in an organization and is a function of position held in the organization. When removed from his or her position in the organization, the person no longer has the authority that is associated with that position. According to acceptance theory of authority, a manager's authority rests with his or her subordinates. The manager has no authority unless subordinates choose to accept his or her commands. A third theory of authority states that the nature of the situation should be the force that grants authority. For example, if a fire starts in a work place, the worker nearest to the telephone might legitimately assume the authority to call the fire department even though he or she had not specifically been directed to do so. A final theory of authority holds that the person most knowledgeable in a given situation has authority by virtue of his or her expertise.

Responsibility is the obligation towards the goals related to the position and the organization.. Managers' primary responsibilities are to examine tasks, problems, or opportunities in relationship to the company's short-and long-range goals. They must be quick to identify areas of potential problems, continually search for solutions, and be alert to new opportunities and ways to take advantage of the best ones.

Delegation

Delegation is the assignment of formal authority and responsibility to another person for carrying specific activities. The delegation of authority by managers to employees is necessary for efficient functioning of any organization because no manager can personally accomplish or completely supervise all of what happens at an organization. The following are guidelines for effective delegation:

- 1) Tasks should be assigned in terms of results expected from a position.
- 2) There should be parity of authority and responsibility.
- 3) There should be well-defined clarification of limits of authority
- 4) Command, orders or guidance should always flow to a sub-ordinate from one delegating superior
- 5) There should be open communication between sub-ordinate and delegating superior

Decentralization and Centralization

Decentralization is the tendency to disperse formal authority. Centralization, on the other hand, is the concentration of authority. In a relatively decentralized organization, considerable authority and accountability are passed down the organizational hierarchy. On the other hand, in a relatively centralized organization, considerable authority and accountability remain at the top of the hierarchy.

Advantages of Decentralization

1. *Unburdening of top managers*
2. *Better decision-making.*
3. *Better training, morale, and initiative at lower levels.*
4. *Promotes development of general managers*
5. *Facilitates product diversification*

6. *More flexibility.*

Disadvantages of Decentralization

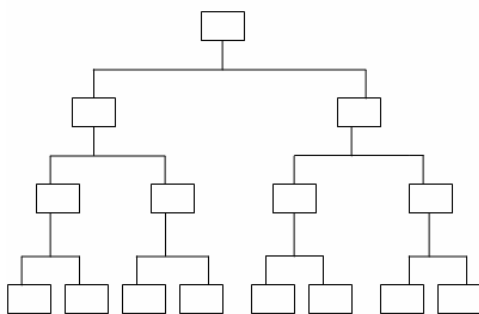
1. *Makes it more difficult to have a uniform policy*
2. *Extensive decentralization may lead to loss of control*
3. *Complexity of co-ordination of organizational units*
4. *Can be limited by the availability of qualified managers*
5. *Decentralization usually entails bringing in additional staff*
6. *More expensive*

Span of Management Control

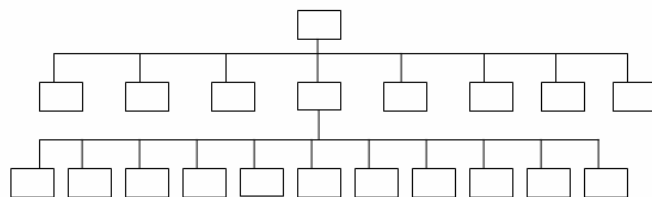
Span of management control (frequently shortened to span of control or span of management) refers to the number of people a manager can effectively supervise.

Choosing an appropriate span of management control for an organizational hierarchy is important for two reasons. First, the span can affect what happens to work relationships in one particular department. Too wide a span may mean that managers are over extended and subordinates receiving too little guidance. When this happens, subordinates may start thinking that they are too remote from the point of control and may become careless. Too narrow span, on the other hand, is inefficient because managers are under utilized.

Second, the span can affect the speed of decision making in situations where multiple levels in the organizational hierarchy are involved. A narrow span of management results in many organizational levels and a long chain of command slows decision-making. In contrast, wide spans results in few organizational levels.



Organization with Narrow Spans



Organization with Wide Spans

Advantages of Narrow Spans

1. *Close supervision*
2. *Close control*
3. *Little or no sub-ordinate training required*

Disadvantages of Narrow Spans

1. Managers under utilized
2. High costs due to many levels of management
3. Excessive distance between lowest level and top level
4. Slow decision-making
5. Superiors tend to get too involved in sub-ordinates' work

Advantages of Wide Spans

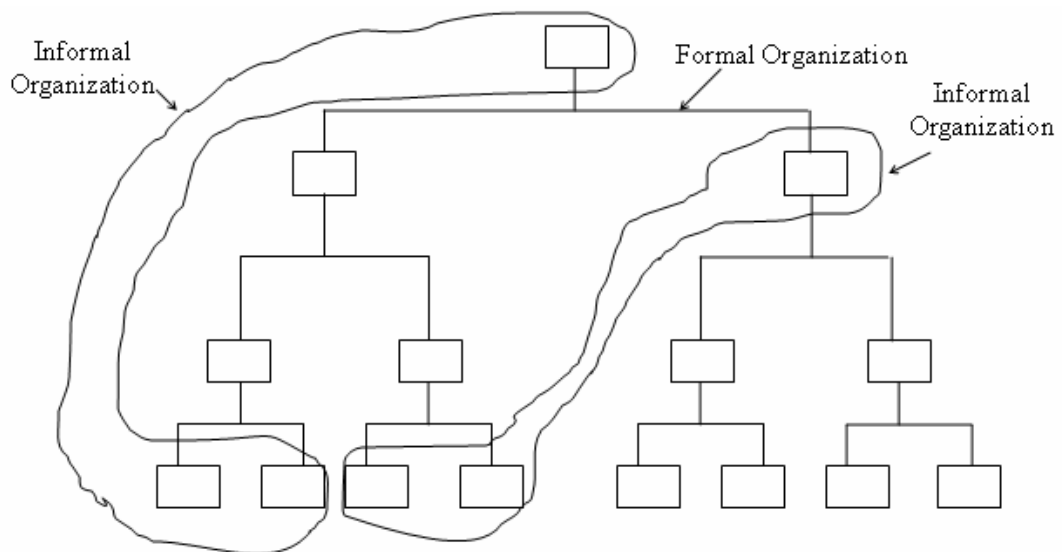
1. Quick decision-making
2. Superiors are forced to delegate
3. Low costs due to few levels of management

Disadvantages of Wide Spans

1. Managers are over extended
2. Requires exceptional quality of managers
3. Tendency of over loaded superiors to become decision bottlenecks

Note: There is no definite number of people a manager can effectively supervise; the number depends on several underlying factors. These include nature of work and capability of managers and sub-ordinates, the degree of sub-ordinate training required and possessed, the clarity of authority delegated, the clarity of objectives, plans and policies, the effectiveness of communication techniques and the type of organization.

Formal and Informal Organization



Formal and Informal Organization

Formal organization means the intentional structure of roles and positions in formally organized enterprise. Informal organization on the other hand is the network of personal

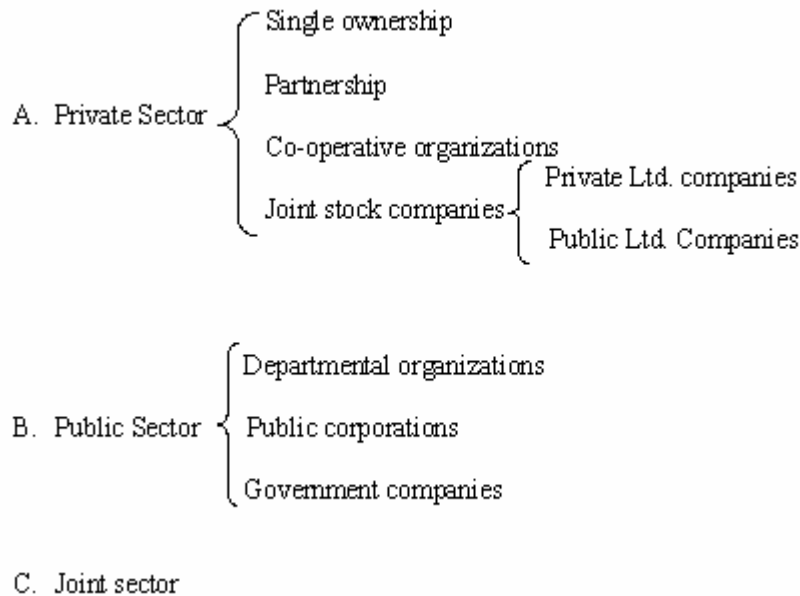
or social relations not established or required by the formal organization but arising spontaneously as people associate with one another.

Business

Business may be defined as *an activity* in which different persons exchange something of value, whether goods or service, for mutual gain or profit.

Forms of business Organizations

The organizational pattern of the firms on the basis of their ownership can be classified as follows:



Private sector

Private sector organizations, as the name indicates, are exclusively owned by private individuals. The efficiency of the private sector organizations is usually very high compared to organizations from any other sector. The important forms of the private sector organizations are:

- 1) Single ownership
- 2) Partnership
- 3) Joint stock companies
- 4) Cooperate organization

Single Ownership

Single ownership is a form of business organization, which is owned and controlled by a single individual. Also known as sole proprietorship, it is the oldest and simplest form of business organization. In this form of organization, an individual introduces his own capital, uses his own skill and intelligence in the management of its affairs and is solely responsible for the results of its operation. The sole proprietor may have any number of persons working for him but they will be just paid employees or friends and relatives having no share in the ownership of the business.

The sole proprietor enjoys full benefit in terms of profit earned by the business. However, he will be personally liable for all kinds of risks attached to his business. His liabilities will be unlimited.

Advantages of Single ownership

- 1) *Ease of formation and dissolution.*
- 2) *Direct relationship between effort and reward serves as a powerful incentive to the proprietor to manage the concern efficiently*
- 3) *Ease of coordination*
- 4) *Promptness in decision-making*
- 5) *Flexibility in management*
- 6) *Secrecy of the affairs of business can be maintained*
- 7) *Freedom from government regulations*

Disadvantages of Single Ownership

- 1) *The amount of capital that can be invested is limited, therefore, rendering it unsuitable for modern business*
- 2) *All the qualities required for success in business are rarely found in a single person*
- 3) *The liability of the sole proprietor will be unlimited*
- 4) *Uncertainty of duration as the firm may cease to exist with the death of the proprietor*

Partnership

Indian partnership act defines partnership as the relation between persons who have agreed to share profits of a business carried on by all or any of them acting for all. The sharing of profits is the basis for defining partnership. The contribution of the partners in running the business need not be same. The minimum number of partners is two and the upper limit is ten for banking business and twenty for general business as per the Indian Companies act. The partnership is created by mutual consent and voluntary agreement. Registration of a business under partnership is essential under shops and establishment act in order to take legal help in enforcing the terms of agreement on the partners. Every partner has an unlimited liability in respect of the firm's debt and limitation of the liability through mutual agreement is not possible legally under partnership.

There is a category of partnership, which is prevalent in western countries known as limited partnership. In this case there are two classes of partners – special and general. The liability of special (or limited) partners is limited to the extent of his investment, and that of general partners is unlimited. In a limited partnership there must be at least one general partner whose liability is unlimited. In India the law does not recognize this type of partnership.

There are many types of partners depending upon their specific role in business. There are *active partners* who bring in capital and take active interest in the conduct of the business. There are *sleeping partners* who bring in capital but do not take active interest in the conduct of the business. Such partners after contributing their share of capital wake up only either to share the profits or to liquidate the business. There are *nominal partners* who lend their reputed name for the company's reputation without

contributing any capital or without any active interest in the conduct of the business. Legally, however, such partners are equally responsible for the liabilities of the firm. There are *secret partners* who bring in capital and take part in the conduct of the business but no where their names appear. There are *minor partners* who are below eighteen years of age and associated with the business. Such partners have limited liability.

Advantages of Partnership

- 1) *Ease of formation as there are very little legal formalities*
- 2) *Larger financial resources as compared to single ownership*
- 3) *Balanced judgment as the partners possesses various sorts of talent, expertise and experience*
- 4) *Adequate credit availability because of unlimited liabilities of the partners*
- 5) *Flexibility of operation*
- 6) *Secrecy in business*
- 7) *Losses, if any, are shared by the partners*

Disadvantages of Partnership

- 1) *Unlimited liabilities of each partner*
- 2) *All partners suffer because of the wrong steps taken by any of the partners*
- 3) *Uncertain life as partnership may dissolve by death or insolvency of a partner*
- 4) *Lack of public confidence as the affairs of the business are kept secret and the accounts is not published*
- 5) *Non-transferability or restricted transferability of the partners' interest in the business*

Joint Stock Company

The joint-stock company is the most important form of business organization. It is a voluntary association of individuals for profit, having a capital divided into transferable shares of different values. A joint stock company is a legal entity with a perpetual succession.

The capital is raised by selling shares of different values and these shares are transferable. Persons who purchase the shares are called *shareholders* and these shareholders elect the managing body known as *board of directors*. The board of directors is responsible for policymaking, important financial and technical decisions and efficient working of the company.

In this form of organization liability of the shareholder is limited to the amount of shares held by him and he is free from the responsibility of the debts and claims of the company beyond the value of shares. Because of this advantage all sections of the people are encouraged to contribute for the company. The shares of a joins-stock company are transferable.

The Joint stock companies are of two main kinds: private limited company and public limited company.

Private Limited Company

A private limited company can be formed by a minimum of two persons and the maximum number of membership is limited to fifty. Transfer of shares is limited to

members only and general public cannot be invited to subscribe the shares. Normally the members of a private limited company are friends and relatives.

A private limited company need not make the prospectus, accounts and other particulars open to public. The members only are entitled to receive a copy of the balance sheet and profit loss accounts yearly along with the auditors report. The government also does not interfere on the working of the company. A private limited company, while conferring the advantage of limited liability, allows a business to be privately owned and managed.

Public Limited Company

A public limited company is one whose membership is open to general public. The minimum number of shareholders required to form such a company is seven, but there is no upper limit.

The public limited companies can advertise to offer its share to general public through a prospectus and there is no restriction on the transfer of shares. These companies are subjected to greater degree of legal control. This control is necessary to protect the interest of the shareholders and the members of the public. The affairs of the public limited company should be made open to public by publishing in leading newspapers.

Advantages of Joint Stock Companies

- 1) *Availability of large capital*
- 2) *Limited liability*
- 3) *Not affected by the death or retirement*
- 4) *Risk of loss is divided among the shareholders*
- 5) *Ease of expansion*
- 6) *Services of specialists can be obtained*
- 7) *Cheaper and better production because of large-scale production with the use of modern technology, which the company can afford*

Disadvantages of Joint-Stock Companies

- 1) *Lack of personal interest on the part of the salaried manager can lead to inefficiency and waste*
- 2) *Board of directors and managers who have intimate knowledge of the financial position of the company may purchase or sell the shares accordingly for their personal profits*
- 3) *Requires a great deal of legal formalities to be observed*
- 4) *Difficult to maintain secrecy*
- 5) *Few shareholders having greater number of shares may secure control over the company*
- 6) *Slow decision-making*

Cooperative Organization or Cooperative Society

A cooperative society is a form of organization where people associate voluntarily and on the basis of equality for the furtherance of their common economic interest. Consumers'

cooperative societies, cooperative credit societies, cooperative farming societies and cooperative housing societies are some examples of this type of organizations.

The primary motive of a cooperative society is to provide maximum service to its members and not to make profits. This does not, however, mean that a co-operative does not work for profit at all. There are several societies engaged in business activities, which earn reasonably good profits while providing service to their members as well as to non-members. Whatever is the profit, it will be partly distributed as bonus to its members.

A cooperative society raises its capital from its members in the form of share capital and a fixed rate of return is paid on the capital subscribed by each members. The shares of a cooperative are not transferable. The management of a cooperative is run by a managing committee elected by members on the basis of *one member one vote* irrespective of the number of shares held by members. The general body of members decides the broad policy framework and guidelines, which the managing committee is required to follow.

A cooperative society is required to be registered under the cooperative societies act. It has a perpetual succession, which is not affected by entry or exit of members.

Advantages of the Cooperative Society

- 1) *Democratic management*
- 2) *Limited liability*
- 3) *The life of a cooperative society is not affected by the death or insolvency of a member*
- 4) *Ease of coordination because of the cooperation among the members of the society*
- 5) *Monetary help can be secured from government*
- 6) *Helps development of moral character*

Disadvantages of the Cooperative Society

- 1) *Limitation of capital*
- 2) *Excessive government regulation*
- 3) *Lack of secrecy*
- 4) *Insufficient motivation*
- 5) *Inefficiency of management as the members generally lacks technical knowledge and may not be competent enough*

Public Sector

Public sector companies are established by the government to produce and supply goods and services required by the society. Public sector prevents the economic unbalance in the nation. It also serves as a means to obstruct the monopolistic tendencies. The important forms of public sector organizations are:

- 1) Departmental organizations
- 2) Public corporations
- 3) Government companies

Public sectors are accountable in terms of their results to parliament and state legislature.

Departmental Organizations

Departmental organizations are organized like any other government departments. A top executive appointed by the ministry concerned will manage the organization. Postal and Telegraphs, Railways, Defense industries, etc. are examples of this type of organization. In certain organizations cooperation from several ministries may be required and in such cases a board or committee of representatives from the ministries concerned will manage the organization.

Public Corporations

A public corporation is usually established by a special act of the parliament or state legislature with internal autonomy. Special statute also prescribes its management pattern, powers, duties and jurisdiction. Though the total capital is provided by the government, they have separate entity and enjoy independence in matters related to appointment, promotions etc. Public service rather than profit maximization becomes the main aim of such corporations. The industrial finance corporation, the Life Insurance Corporation, etc are examples of public corporations.

Government Companies

A government company, according to the Indian companies act, 1956, is any company in which not less than 51% of the share capital is owned by the Central Government or by any state government or governments, or partly by the Central government and partly by one or more state governments. It is organized in the form of a joint stock company. Hindustan machine tools Ltd., Hindustan Aircrafts Ltd., Hindustan shipyard Ltd., etc are examples of this type which are owned by the government and also be joint stock companies.

These companies are managed by elected board of directors. In its day-to-day working it is free from government interference. However, bureau of public enterprises can issue guidance and directions.

Advantages of Public Sector

- 1) *Helps for the betterment of the community and for the welfare of the people*
- 2) *Facilities like power, transport, credit, and insurance, etc are easily made available to public sector units*
- 3) *Because of the government control economical and social objectives can easily be achieved*
- 4) *Provides better working conditions to the employees and cheaper and better products and services to the customers*
- 5) *Encourages industrial growth of under-developed regions in the country*
- 6) *Provides employment opportunities to all sections of the people*
- 7) *Prevents monopolistic tendencies and paves the way for equitable distribution of wealth among different sections of the community*

Disadvantages of Public Sector

- 1) *Because of bureaucratic control generally timely decisions are not taken*

- 2) *Lack of initiative among workers because promotions are seniority based rather than merit based*
- 3) *Too much of interference by the political leaders and government in the internal affairs of public sector units*
- 4) *Misuse of excessive freedom (compared to private concerns) cannot be ruled out*
- 5) *Inadequate accountability*
- 6) *Government officials prefer to work according to certain rules and regulations and therefore lesser flexibility*
- 7) *Incompetent persons may occupy high levels*

Joint Sector

The concept of joint-sector implies the participation of both the government and the private sector in the share capital and general management of the business. It combines the best aspects of both private sector and public sector organizations and aims at achieving the task of social justice through efficient use of resources. Joint sector firms can be a pure Indian firm or an Indian firm with foreign collaboration

Advantages of Joint Sector

- 1) *Helps to foster the industrial development with social justice*
- 2) *Checks business malpractices*
- 3) *Antidote to monopoly and concentration of economic power*
- 4) *Combines the best aspects of both private sector and public sector organizations.*
- 5) *Makes nationalization unnecessary*

Limitations of Joint Sector

- 1) *Lack of confidence between two sectors*
- 2) *Managerial autonomy making the owners passive in business*
- 3) *Inadequate accountability*

Module 2

Management

Management is the process of creating an atmosphere, wherein individuals working together in groups, accomplish a given objective with the highest degree of productivity.

Management is the process through which people are mobilized to achieve designated goals.

Importance of management

- 1) Management helps in increasing the effectiveness and efficiency and thereby productivity of the enterprise as well as the individual worker.
- 2) Management helps in development of full human potential.
- 3) Management helps in raising the worker morale.
- 4) Management helps in building mutual trust.
- 5) Management helps developing teamwork.
- 6) Management helps in providing a stable livelihood for all employees.
- 7) Management helps in constantly and forever improving the system of production and service.

Characteristics of Management

- 1) Management applies to any kind of organization.
- 2) Management applies to managers at all organizational levels.
- 3) The aim of all managers is the same: to be productive.
- 4) Managing as a practice is an art in which practitioners apply the underlying theory and science in light of situations.
- 5) Management attempts to create a desirable future, keeping the past and present in mind.
- 6) There are various approaches to management.

Levels of management

The term “Levels of Management” refers to a line of demarcation between various managerial positions in an organization. The number of levels in management increases when the size of the business and work force increases and vice versa. The level of management determines the chain of command, the amount of authority, and status enjoyed by any managerial position. Although it would be possible to slice the management structure in an organizational hierarchy into any number of vertical levels usually three levels are cited namely:

- 1) Top management
- 2) Middle management, and
- 3) Supervisory or first level management

Managers at all these levels perform different functions. The role of managers at all the three levels is discussed below:

Top level Management

Top level management is responsible for framing policies of the organization. All critical decisions are also made at this level. Top level management consists of board of directors, managing director, general manager and senior most managers. Top level management is administrative in nature. Following are the important functions performed by top level management:

- 1) Develops long - range plans and strategies
- 2) Top management lays down the objectives and broad policies of the enterprise.
- 3) Issues necessary instructions for preparation of department budgets
- 4) Consults subordinate managers on subjects or problems of general scope
- 5) Involved in selection of key personnel
- 6) Controls and coordinates the activities of all the departments

Middle level Management

Middle level management is the link between top level and low level management. They devote more time to organizational and directional functions. These managers supervise, direct and control the activities of foremen, inspectors and supervisors. The activities at this level include:

- 1) Makes plan of intermediate range and prepares long - range plans for review by top level management
- 2) Establishes departmental policies
- 3) Counsels subordinates on production, personal or other problems
- 4) Selection and recruitment of personnel
- 5) Training of lower level management
- 6) Interpret and explain policies from top level management to lower level
- 7) Coordinating activities within the division or department
- 8) Sends important reports and other relevant data to top level management
- 9) Evaluate performance of junior managers
- 10) Motivate lower level managers towards better performance

Lower level Management

Lower level is also known as supervisory / operative level of management. It consists of foremen, inspectors, supervisors etc. They will be mainly concerned with direction of operative employees and the major functions performed at this level include:

- 1) Makes detailed, short - range operating plans
- 2) Assigning of jobs and tasks to various workers
- 3) Supervise and guide the sub-ordinates
- 4) Reviews performance of subordinates
- 5) Supervises day - to - day operations
- 6) Responsible for the quality as well as quantity of production
- 7) Communicate workers problems, suggestions, and recommendatory appeals etc. to the higher level and higher level goals and objectives to the workers
- 8) Solve the grievances of the workers.

- 9) Training of workers
- 10) Arrange necessary materials, machines, tools etc. for day - to - day operations
- 11) Prepare periodical reports about the performance of the workers
- 12) Ensure discipline in the enterprise
- 13) Motivate workers

Managerial Skills

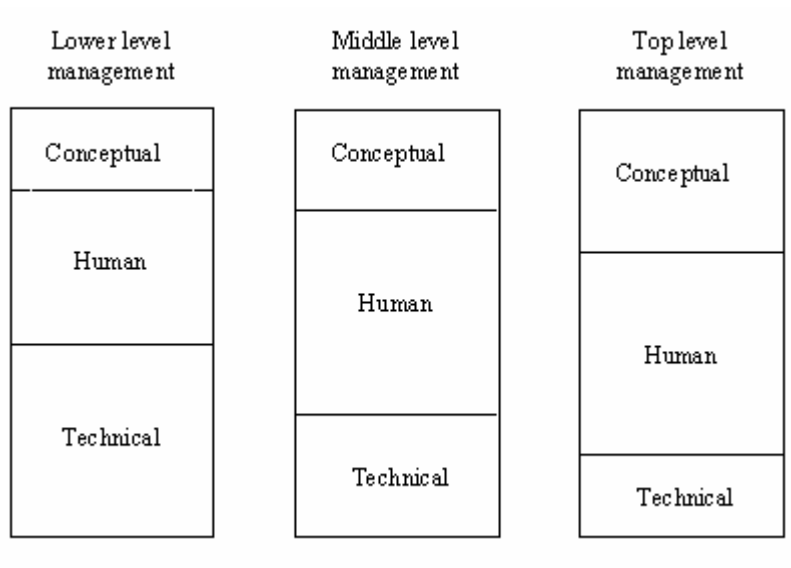
Despite variations in the duties and responsibilities of a manager, there are several skills that all managers must develop. Three basic and essential skills that are needed by all managers are technical, human, and conceptual.

Technical skill is the ability to perform a manager's job. An accountant, doctor, engineer or a musician all have technical skills in their respective fields of specialization. A manager must possess technical skill.

Human skill is the ability to work with others by getting along with them, motivating them, and communicating effectively with others. The manager must focus his attention on improving his interpersonal relations with peers, subordinates, and his own supervisors.

Conceptual skill is the ability to coordinate and integrate the entire organizational interests and activities. A manager must have the ability to see the organization as a whole and not make decisions from his own departmental point of view. He must be able to see how his department is affected by the decisions of others.

The relative importance of skills varies according to the level of management as illustrated in the following figure.



Skills needed at different levels

Scientific Management and Contributions of Early Thinkers

Scientific management is a theory of management that analyzes and synthesizes workflows, with the objective of improving labor productivity. The core ideas of the theory were developed by Frederick Winslow Taylor (1856 – 1915) in the 1880s and 1890s. Taylor believed that decisions based upon tradition and rules of thumb should be replaced by precise procedures developed after careful study of an individual at work. Scientific management's application is contingent on a high level of managerial control over employee work practices.

The contributions of early thinkers towards development of management theory are detailed below:

F.W. Taylor

F. W. Taylor (1856 – 1915) known as father of scientific management rested his philosophy on four basic principles:

1. Develop a science for each element of a man's work, which replaces the old rule of thumb method.
2. Scientifically select and then train, teach, and develop the work man, where as in the past he chose his own work and trained himself as best as he could.
3. Heartily cooperate with the men so as to ensure all of the work being done is in accordance with the principles of science, which has been developed.
4. There is an almost equal division of the work and responsibility between management and workmen. The management takes over all work for which they are better fitted than the workmen, while in the past almost all of the work and the greater part of the responsibility were thrown upon the men.

Taylor believed that management and labour had a common interest in increasing productivity and the success of these principles required “a complete mental revolution” on the part of management and labour.

Taylor stressed the importance of “time and motion study” to increase efficiency of men and machines. He introduced a wage incentive plan known as differential rate system, which involves payment of higher wages to more efficient workers.

Henry L Gantt

Henry L Gantt (1861-1919) worked with Taylor on several projects and was his close associate. He improved upon Taylor's differential piece rate system and came up with a new idea. Every worker who finished a day's assigned workload would win a 50% bonus. The supervisor would also earn a bonus for each worker who reached the daily standard, plus an extra bonus if all the workers reached it. This would motivate the supervisors to train their workers to do a better job.

Gantt also devised a charting system for production scheduling, now known as Gantt chart. The Gantt chart is still in use today. It also formed the basis for two charting

devices which were developed to assist in planning, managing and controlling complex organizations: the Critical path Method (CPM) and Program Evaluation and Review Technique (PERT).

The Gilbreths

Frank B and Lillian M Gilbreth (1868-1924 and 1878-1972) made their contribution to the scientific management movement as a husband and wife team. They did a lot of research in order to improve work methods and thus to discover one best way of accomplishing a task. Their main field of interest was fatigue and motion studies and focused on ways of promoting the individual worker's welfare. To them, the ultimate aim of scientific management was to help workers reach their full potential as human beings.

In their conception, motion and fatigue were intertwined – every motion that was eliminated reduced fatigue. Using motion picture cameras, they tried to find the most economical ways of doing jobs. They concluded that fatigue could be considerably reduced by lightening the load, spacing the work and by introducing rest periods.

Frank Gilbreth published a series of books describing the best way of laying bricks, handling materials, training apprentices, and improving methods while lowering costs and paying higher wages.

Dr. Lillian Gilbreth is often known as the first lady of management. Lillian's thesis-turned-book, *The Psychology of Management*, is one of the earliest contributions to understanding the human side of management.

Lillian faced many incidents of discrimination during her life, including the fact that her book could only be published if her initials were used so readers would not know she was a woman. Her work illustrated concern for the worker and attempted to show how scientific management would benefit the individual worker, as well as the organization. Lillian wrote about reduction of worker fatigue, how to retool for disabled veteran workers returning to the workplace, and how to apply principles of scientific management to the home.

Harrington Emerson

Harrington Emerson (1853–1931) was focused on development of principles of efficiency. The Engineering Magazine published a series of articles by Emerson in 1908 and 1909 that were later issued as a single volume. In 1913, Emerson published *Twelve Principles of Efficiency*. This publication became a landmark in the history of management thought.

Emerson was also a strong advocate of making a strict distinction between line and staff roles in organizations. Emerson embraced the general staff concept where each firm was to have a chief of staff and four major sub groupings of staff under him: one for employees, one for machines, one for materials, and one for methods. Staff advice was

available to all levels and focused on planning. Moreover, Emerson urged on the use of statements of goals and objectives for the total organization.

Emerson made other contributions in the areas of cost accounting and in setting standards for judging workers and shop efficiency.

Principles of Management as Laid Down by Henri Fayol

Henri Fayol (1841-1925) is known as the Father of principles of management. Fayol believed and prescribed fourteen principles that would aid in setting up and managing organizations. These principles are listed below.

1. Division of work

Work must be divided into tasks, sub-tasks and still smaller units till specialization is achieved.

2. Authority and responsibility

A relationship must be established between the responsibility and the authority a manager exercises. If a subordinate is given responsibility, he should also be given authority to go with it.

3. Discipline

This principle deals with the sanction of rewards for good work or meeting standards and punishment for poor work or failure to meet standards.

4. Unity of Command

Each employee must receive instructions from only one person. Fayol believed that when an employee reported to more than one manager, conflicts in instructions and confusion of authority would result.

5. Unity of direction

Tasks must be regrouped by departmentalization under one head whose major responsibility is coordinating activities.

6. Subordination to general interest

This principle is based on the idea that the whole is greater than the sum of its parts. General interest supercedes the interests of individuals.

7. Remuneration

Compensation for work done should be fair to both employees and employers.

8. Centralization

Decreasing the role of subordinates in decision-making is centralization; increasing their role is decentralization. An optimal balance between centralization and decentralization exists for each situation. This balance must be determined by taking the manager's capabilities into consideration.

9. Scalar chain

This refers to a graded chain of managers from ultimate authority at the top to lowest ranks, resulting in hierarchical levels. This principle also states that authority and responsibility should flow in a direct line vertically from top to bottom.

10. Order

This principle emphasizes the importance of arranging and organizing human and physical resources logically and neatly.

11. Equity

Managers should be both friendly and fair to their subordinates.

12. Stability

In order to provide stability of an organization, long-term commitments must be encouraged.

13. Initiative

Employees must be encouraged to think through and implement a plan of action.

14. Unity of effort

Coordination and unity are important to achieve the goals of an organization. To achieve unity and coordination communication is essential.

Functions of Management

The managerial functions provide a useful framework for organizing management knowledge. Managerial functions can be basically grouped under planning, organizing, motivating, controlling, coordinating and decision-making.

Planning

Plans give the organization its objectives and set up the best procedures for reaching them. Plans made by top-level management may cover periods as long as five or ten years. On the other hand, the middle and lower level managers focus on short-range and day-to-day plans. The elements included in the planning function are:

- 1) The policies that will help to achieve objectives.
- 2) The programmes that a manager will carry out
- 3) The time schedules that a manager will have to meet
- 4) The budgetary considerations that will be involved

All the above elements are equally important and interact with all other elements

Organizing

Organizing is the process of arranging and allocating work, authority, and resources among an organization's members so that they can achieve organization's goals. The elements included in organizing function are:

- 1) Grouping of activities necessary to accomplish organization's goals in the light of the human and material resources available and the best way, under the circumstances, of using them.
- 2) Delegating to the head of each group the authority necessary to perform the activities.
- 3) Establishing relationships that will provide each with the necessary information.
- 4) Scrutinizing the relationships between various units and the effect of operation of these units on each other.

Organizing is a never-ending process. All types of organizations are in a continual state of being reorganized. When goals and programmes are redirected, activities also change.

Sometimes staffing function is considered as a part of organizing function. Staffing is the function of manning the organization structure and keeping it manned. The main purpose of staffing is to put right man on right job. Staffing involves:

- 1) Manpower Planning (estimating man power in terms of searching, choose the person and giving the right place)
- 2) Recruitment, selection & placement
- 3) Training & development.
- 4) Remuneration
- 5) Performance appraisal
- 6) Promotions & transfer

Motivating

Motivation is a human psychological characteristic. It pertains to various drives, desires, needs, wishes and other forces.

Motivation is not easy to achieve and what a manager can try to do is to create a working climate in which all members may contribute to the limits of their ability. The key elements in such a work situation and its effect on the employee are known to be:

- 1) The degree to which the employee feels his goals and those of the organization are similar.
- 2) The employee's relationships with his coworkers and especially with his supervisor.
- 3) The way in which his job helps him meet his needs for present income and future security and does so in a manner that seems fair.
- 4) The extend to which it enables him to feel adequate to his tasks and to gain a sense of accomplishment for jobs well done.

Motivational function provides a great deal of challenge to a manager. He must have the ability to identify the needs of his subordinates and the methods and techniques to satisfy those needs. Motivation is a continuous process as new needs and expectations emerge.

Controlling

Controlling is the process of ensuring that actual activities conform to plan activities. Through the controlling function, the manager can keep the organization on the right track before it deviates too far from its goals. The controlling function involves:

- 1) Establishing standards of performance.
- 2) Measuring actual performance.
- 3) Comparing actual performance to the established standards.
- 4) Taking corrective action if deviations are detected.

For the control to be effective, a system of communications or reports is required to inform the manager of the facts on which to base measurements, comparisons and corrective action. A great deal of the manager's time is involved in controlling.

Coordinating

Coordination is the process of integration of the activities of separate departments of an organization to accomplish organizational goals. Coordination is needed both up and down the organization structure and laterally as well. It can also occur among people working at different organizations. The extent of coordination depends on the nature of activities performed and the type of organization structure.

Some authors consider coordinating as a part of organizing function as organizing involves a great deal of coordinating effort.

Decision-making

Decision-making is the process of identifying and selecting a course of action from among alternatives. Decision-making is an important part of every manager's job and it requires all the skill and judgment a manager accumulates over the years.

The manager constantly seeks to make correct decisions involving the use of the various types of resources at his disposal to attain the various objectives. A manager decides on the utilization of men, materials and machines to achieve such goals as quality, low cost, quick delivery, safety and so on.

Directing

Directing is said to be a process in which the managers instruct, guide and oversee the performance of the workers to achieve predetermined goals. Directing is said to be the heart of management process. Planning, organizing, staffing has got no importance if direction does not take place.

Directing initiates action and it is from here actual work starts. Directing consists of process or technique by which instruction can be issued and operations can be carried out as originally planned. Therefore, Directing is, therefore, guiding, inspiring, overseeing and instructing people towards accomplishment of organizational goals.

Directing concerns the total manner in which a manager influences actions of subordinates. It includes the following elements:

- 1) Issuing orders that are clear, complete and within the capabilities of subordinates to accomplish.
- 2) Suggesting an incessant training activity in which subordinates are given instructions to enable them to carry out the particular assignment in the existing situation.
- 3) Motivating the workers to meet the expectations of the manager
- 4) Maintaining discipline and rewarding those who perform well

Module 3

Marketing Management

Marketing management is one of the main management disciplines, encompassing all the strategic planning, operations, activities, and processes involved in achieving organizational goals by delivering value to customers. Marketing management focuses on satisfying customer requirements by identifying needs and wants, and developing products and services to meet them.

Concept of Marketing Versus Selling

The marketing concept gives importance to consumers and the concept holds that the key to achieving organizational goal consists in determining the needs and wants of customers and delivering the desired satisfaction.

The selling concept is product-oriented and the concept holds that the consumers, if left alone, normally will not buy enough of the organizations' products and therefore efforts must be made to force the customers to buy the products.

Marketing focuses on the needs of the buyer, whereas selling focuses on the needs of the seller. That is, marketing is consumer oriented, where as selling is product oriented.

In marketing primary attention is placed on consumer needs and wants at the planning and development stage of the product, where as in selling efforts are only to sell what has been produced without taking into consideration the consumer preferences

Marketing Mix

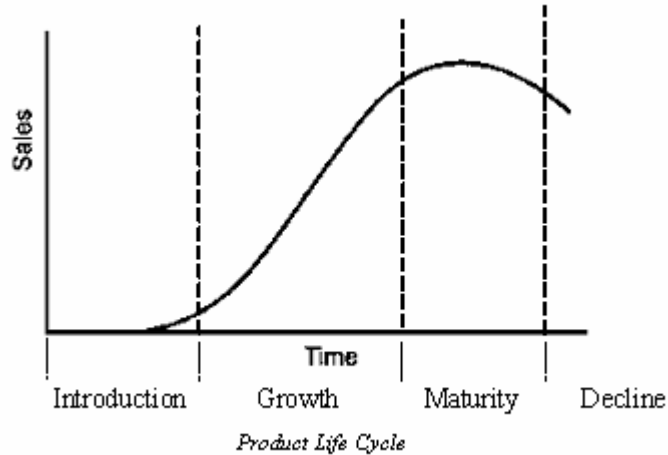
The marketing mix refers to the blend of four controllable variables that companies use to influence the market responses – product, pricing, promotion and channels of distribution. Marketing mixes used by various organizations may vary depending upon the strategies and tactics adopted by them.

Product

Prior to the adoption of the marketing concept philosophy the major focus was on the technological or engineering dimensions of a product. However, today products are designed and developed by taking into consideration the needs and demands of the customers. From the marketing view point an optimal product design would be one that will bring enough profits to the company and at the same time meets the needs and expectations of the customer. Ideally speaking, the product should be an innovative one and should be available in any desired quantities at an attractive cost. Whether the customers will purchase the product again and whether they will develop favourable opinion about the firm will depend upon the aesthetic and functional qualities of the product. These qualities that will influence the customers are subjects of market research and will have a bearing on matters related with choice of product design. The selection of one design over another may lead to higher product prices and also may require higher promotional expenditures. As the decisions regarding production and promotional

activities cannot be frequently changed from time to time, any change in design should be properly planned and thought off before implementation.

Product Life Cycle



A product life cycle refers to a series of successive stages a product undergoes from the time it is put in the market till it is withdrawn from the market. In the early stage when the product is introduced (referred to as the introductory stage) the level of sales will be very low. This stage is followed by a period of rapid growth in sales (called as the growth stage). At this point the product reaches the maturity stage when the sales have more or less peaked. Finally, the curve slopes downwards and this is called the declining stage.

(While in general, this pattern is common, it may vary from product to product. Moreover, a product may jump a stage or two, or remain stagnant in a particular stage. For example, a fashion may disappear soon after a boom, but a product like salt has come to stay forever.)

Pricing

Price refers to the exchange value of a product or a service and one of the most important tasks of a marketing manager is pricing. Unless the product is priced properly, all other elements of marketing mix may be rendered ineffective. The key factors to be considered in pricing are:

- 1) Nature and extent of consumer demand
- 2) Total cost associated with the product
- 3) Legal aspects
- 4) Prices for similar products offered by competing firms
- 5) General economic conditions
- 6) Type of customers
- 7) Method of promotion
- 8) Distribution channels
- 9) Life cycle of the product
- 10) Profit considerations

Pricing Methods

The two common methods of pricing are (i) the cost plus approach and (ii) the market or demand approach

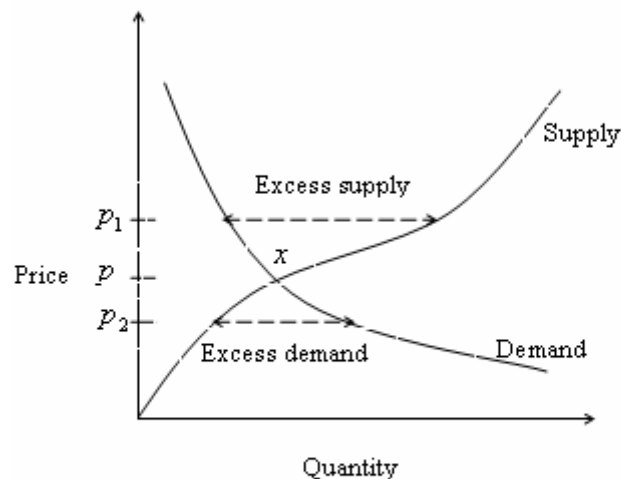
Cost Plus Approach

Cost plus approach is one of the commonly used methods of pricing. In this method total cost associated with production, selling and distribution are added with the desired profit to arrive at the product price. That is,

$$\text{Product Price} = \text{Fixed Cost per Unit} + \text{Variable Cost per Unit} + \text{Profit margin}$$

The Market or Demand Approach

In this method of pricing, prices are not based on the total cost but allow the market forces (supply and demand) to determine the product price. The supply and demand curves shown below indicate that with the increase in price the demand for the product decreases. Similarly, as the prices fall demand will increase. Usually there is a maximum that a demand can reach no matter how the price falls. Also it is evident that when the prices are low there are a very few suppliers and the quantity available is low. Similarly, when the prices are high the number of suppliers will go up leading to increased quantity of goods.

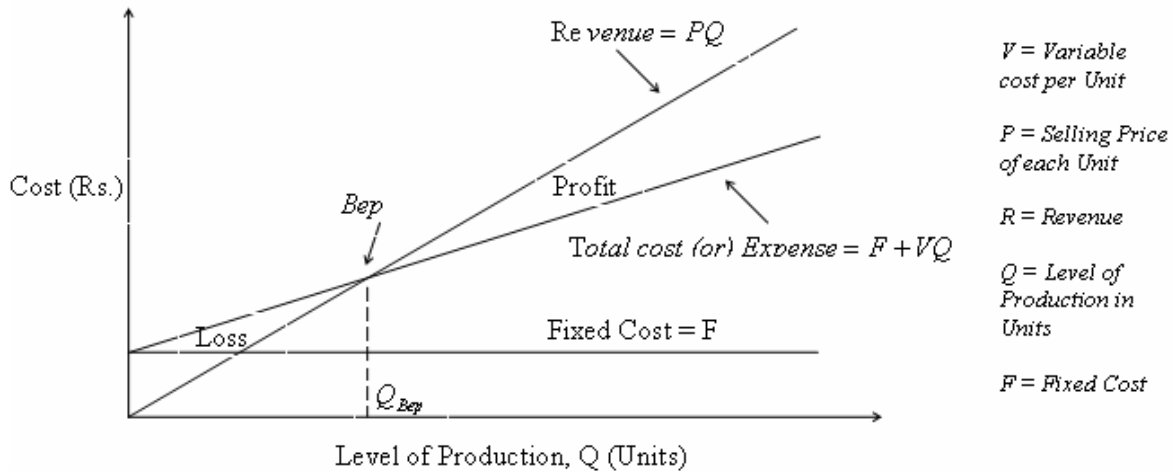


Market forces determining the product price

With the supply and demand curves as shown in the above figure the market will eventually settle at price p corresponding to point x , where supply and demand match. Below the price p , such as p_2 , there is an excess of demand over supply and the prices will be pushed upwards towards point x , as the consumers compete with each other to obtain limited stock. Above the price p , such as p_1 , there is an excess of supply over demand, resulting in excess stock and the suppliers will be forced to bring down the price towards point x , so as to sell out the excess unsold items.

However, in utilizing this method the marketing manager should distinguish between market demand and company demand. Market demand refers to the total amount of purchase potential for a type of product, whereas company demand refers to the purchase potential for the company's product (of the same type).

Break even Analysis



Fixed Costs

Land
Building
Administrative cost
Staff salaries
Insurance

Variable Costs

Material Cost
Labour cost
Spare parts cost
Depreciation

Break Even Chart

Any production activity consists of fixed cost F in the form of land, building, equipment etc. which is totally independent of volume of production. This cost is always accompanied with a variable cost which roughly varies in a direct proportion with Q (production level) and total cost is the sum of these 2 components. The break-even point corresponds to the production level at which the firm neither incurs a loss nor enjoys a profit.

At break even point,

$$\text{Total cost (or) Expense} = \text{Revenue}$$

$$\text{That is, } F + VQ_{Bep} = PQ_{Bep}$$

$$\Rightarrow Q_{Bep} = \frac{F}{P - V} \quad (\text{Units})$$

Always efforts must be made to keep the break-even production level as low as possible and this can be achieved by 3 methods, namely

- 1) Reducing fixed cost, F
- 2) Increasing selling price, P and,
- 3) Decreasing the variable cost, V.

Problem

The following data pertains to XYZ Company.

Fixed cost for the year 2009- 2010 = ₹800000

Variable cost per unit = ₹ 40

Selling price of each unit = ₹200

- a) Find the break even point
- b) If the likely sales turnover for the next budget period is ₹ 1600000. Calculate the estimated profit.
- c) If the profit target of ₹600000 has been budgeted, compute the sales turnover required.

Solution

$$\text{a) } Q_{Bep} = \frac{F}{P - V} = \frac{800000}{200 - 40} = 5000 \text{ Units}$$

Break-even point in terms of units is 5000 units and the corresponding money value is $5000 \times 200 = ₹ 1000000$

$$\text{b) Sales turn over (or revenue) } = PQ = ₹1600000$$

$$\Rightarrow Q = \frac{1600000}{200} = 8000 \text{ Units}$$

$$\text{Total cost} = \text{Fixed Cost} + \text{Variable Cost} = F + VQ = 800000 + 40 \times 8000 = ₹ 1120000$$

$$\text{Therefore, Profit} = 1600000 - 1120000 = ₹ 480000$$

$$\text{c) Profit} = \text{Revenue} - \text{expense} = PQ - (F + VQ)$$

$$\Rightarrow \text{Profit} = Q(P - V) - F$$

$$\Rightarrow Q = \frac{(\text{Profit} + F)}{P - V}$$

$$\Rightarrow Q = \frac{1400000}{160} = 8750 \text{ Units}$$

$$\therefore \text{Sales turnover} = 8750 \times 200 = ₹ 1750000$$

Problem

A company has an installed capacity to produce 20,000 units of a certain product. The fixed cost is ₹15 lakhs and the variable cost is ₹100 per unit. What should be the selling

price if it is to break-even at 50 % capacity? What is the maximum profit that can be earned in a year? What will be the profit at 90 % capacity utilization?

Solution

Taking $F = 15$ lakhs, $V = ₹100$ and $Q_{\text{bep}} = 10000$ units, calculate P .

Using the calculated value of P , $F = 15$ lakhs, $V = ₹100$ and $Q = 20000$ units, calculate the maximum profit.

Repeat the step (ii) by taking $Q = 18,000$ units in place of 20000 and calculate the profit.

Promotion

Promotion is communication to the potential buyer and includes all those activities that will facilitate the sale of a good or service or acceptance of an idea. Three important types of promotion are advertising, personnel selling and sales promotion.

Advertising

In practice, product design is a result of some sort of compromise between infinite variety on one hand and the designer's concept of the ideal design on the other. In order to try selling this compromise to potential customers, management resorts to an advertising campaign the policy of which is dependent on the characteristics of the "compromised design" and on how far it conforms or differs from, the expressed desires of the market to which such a campaign is directed. Generally, the main objective of advertising is to expand the market, this being achieved by:

- 1) Providing general information about the existence of the product.
- 2) Providing technical information about its functional characteristics or utilitarian purposes.
- 3) Drawing the customers' attention to those attributes of the product which he wants.
- 4) Winning undecided customers by exhibiting possible attractions (such as color, design, novelty and price) that may persuade him to prefer the product to one offered by competitors.
- 5) Creating demand among a passive population of customers.
- 6) Educating the customer, or telling him what he should want.

Apart from these direct techniques, management may have some additional aims such as increasing the prestige of the firm as a whole, banking on the popularity of one product to strengthen or introduce another or to publicize one aspect of the firm's activity for the purpose of raising money or deviating attention from other activities and so on. Once the design features of a product have been ascertained, appropriate advertising methods can be selected.

Advertising Medium

An advertising medium is a carrier of information to be advertised. The commonly used media for advertising are as follows:

- 1) Newspapers
- 2) Magazines
- 3) Catalogues and leaflets
- 4) Radio

- 5) Television
- 6) Internet
- 7) Public transports such as buses, trains, etc.
- 8) Outdoor billboards (hoardings) such as on highways
- 9) Slides and film shows in cinemas

Personal Selling

Personal selling refers to oral presentation to one or more customers in a face to face situation to persuade them to purchase the product. It involves two way communications between salesman and buyers. Some of the positive aspects associated with personal selling are as follows:

- 1) The salesman act as a consultant to the customers
- 2) The salesman can adjust his message and presentation according to the needs of the situation and type of customer
- 3) The salesman can clear the doubts of the customers related with the use or functional aspects of the product
- 4) Personal selling develops social relations with customers and builds reputation of the company
- 5) Quick feed back can be obtained through personnel selling

Sales Promotion

All the activities that go into the development of sales or those that are intended to raise the demand level for a product very quickly can be grouped under the title sales promotion. The whole idea behind sales promotion is to bring the name of product and that of the manufacturer constantly before wholesalers, retailers and the consumers in order to stimulate the interest in the product.

Methods of Sales Promotion

Sales promotion can be achieved by resorting to the following:

- 1) *Consumer Promotion*: Persuading consumers to buy; these include samples, money refund offers, prices-off, trading stamps, contest and competitions, etc.
- 2) *Trade Promotion*: Giving incentives to distributors and others to hold stocks of company products. These incentives include special discounts, one or two free units per bulk container, dealer competitions etc.
- 3) *Sales Force Promotion*: Offering bonuses, contests etc. for the salesmen.
- 4) *Good Public Relations*: Developing goodwill among general public and boosting sales. Every proposed business policy should first be analyzed in terms of its effect upon the company image.
- 5) *Display*: Displays at the point of sale, using posters, banners, placards and leaflets, to attract the customers' attention to the product.
- 6) *Good Customer Relations*: Good customer relations are basically the result of the past transactions with the company. Speedy handling of complaint, assistance in emergencies, abiding by announced policies, etc. all develops good customer relations and increase future sale of company products.
- 7) *Product exhibitions, demonstrations, and conferences.*

- 8) *Latest product styling and appealing product packaging catch the eye of the consumer and develop sales volume.*

Channels of Distribution

A channel of distribution is the route that the product follows in its passage from the producer to the consumer. Critical factors affecting this route are:

- 1) Nature of good
- 2) Nature and location of the markets
- 3) Price of the product
- 4) Availability of middlemen, transportation
- 5) Sales effort required by the middlemen, and
- 6) Resources of the manufacturer

Channel management mainly deals with two problems:

- 1) Channel selection, and
- 2) Maintaining the channel

Three policy alternatives may be considered in the selection of the channel:

- 1) The policy of general or intensive distribution whereby the firm seeks to obtain the widest possible distribution for its product by allowing it to be sold everywhere by anyone willing to stock it.
- 2) The policy of selective distribution, where the manufacturer chooses only those outlets that are best able to serve that company's needs.
- 3) The policy of using exclusive dealerships, which allows only one distributor to stock and sell the product in a given market.

A company may use more than one channel, particularly if its market is diversified.

The second problem of channel management is maintaining the channel so that no blockages develop that can adversely affect the company's competitive and profit position. If a firm is selling part of its product by means of its own sales force, the channel maintenance problem is one of maintaining fair and consistent policies. When other type of middlemen are employed (such as wholesalers and agents), the marketing manager will have to focus on three aspects:

- 1) Ensure that the terms of contract are followed by all parties.
- 2) Maintain good relations with the middlemen and encourage them to cooperate fully.
- 3) Represent his company to the middlemen by seeing that sufficient cooperation and assurance are provided.

Marketing Research

Marketing research is the systematic gathering and analysis of data relevant to any problems in the field of marketing.

Market Research Techniques

- a) *Desk Research* - Research is done by summarizing published sources and is a form of secondary research. Analysis is done on past sales, sales fluctuations,

- sales and promotional expenditures, economics of order size, profit per rupee invested etc.
- b) *Surveys* - With concise and straightforward questionnaires, a sample group that represents the target market can be analyzed. The larger the sample, the more reliable results will be. The surveys can be of different types as discussed below :
 - i. In-person surveys are one-on-one interviews typically conducted in high-traffic locations such as shopping malls. In-person surveys can generate response rates of more than 90 percent, but they are costly.
 - ii. Telephone surveys are less expensive than in-person surveys but, due to consumer resistance to relentless telemarketing, convincing people to participate in telephone surveys has grown increasingly difficult. Telephone surveys generally yield response rates of 50 to 60 percent.
 - iii. Online surveys usually generate unpredictable response rates and unreliable data, because there is no control over the pool of respondents. But an online survey is a simple and inexpensive way to gather customer opinions and preferences.
 - iv. Mail surveys are a relatively inexpensive way to reach a broad audience. They are much cheaper than in-person and phone surveys, but they only generate response rates of 3 percent to 15 percent.
 - c) *Depth interview* - A type of qualitative research, which involves long, probing interviews without the use of a formal questionnaire.
 - d) *Observation* - A research technique in which no direct questions are asked, but people in a public place are watched and their behaviour recorded.
 - e) *Hall test* - Getting a group of people (*typically 50*) together in a public hall, usually to see a product demonstration and to fill in questionnaires on the spot.
 - f) *Statistical Methods* - Statistical methods make use of large pre collected data and logically conclude the market investigations.

Sales Forecasting

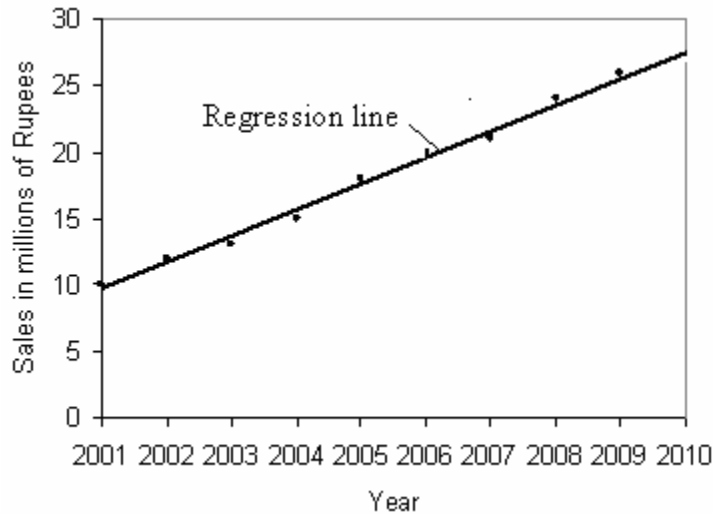
Sales Forecasting is the process of estimating what your business's sales are going to be in the future.

The following are the different methods of sales forecasting.

- 1) *Historic estimate*: This technique makes use of the assumption that what happened in the past will happen in future. For example, if a concern has sold 5000 blankets in winter last year, it will be able to sell the same quantity in winter this year also. Historic estimate is useful if the activity is affected by pattern of seasonality. Historic estimate is not scientifically valid and thus it is not an accurate method; the total sales forecast provided by this method should be modified by other techniques.
- 2) *Sales force estimate*: The sales force method is a sales forecasting technique that predicts future sales by analyzing the opinions of sales people as a group. Sales people continually interact with customers, and from this interaction they usually develop a knack for predicting future sales. The sales force estimation method is considered very valuable management tool and is commonly used in business and industry throughout the world. This

method can be further improved by providing sales people with sufficient time to forecast and offering incentives for accurate forecasts.

- 3) *Trend Line Technique*: Trend Line Technique is employed when there is an appreciable amount of historical data. This technique involves plotting historical data between sales on Y-axis and time on X-axis. A single best fitting line is drawn and projected to show sales estimate for the future.



Trend Line Method

- 4) *Correlation Technique*: This technique makes use of cause-and-effect relationship between sales and some other phenomena that are related to sales. This technique is employed when an organization finds that the sale of its product has a remarkable relationship with the sales of a leading product of another organization. For example, sales of automobile replacement batteries can be correlated with the sale of new cars.
- 5) *Market Survey*: This technique finds application when a concern introduces a new product in the market and is interested to estimate its sales forecast. For a new product, naturally, no historic or past data regarding sales will be available. This technique may be very informal, utilizing the sales force to feel out the potential customers in order to establish the extent of the market or it may be a systematically conducted survey using special mathematical tools. Generally, the new product will be introduced in a relatively small critical trial area, market reaction is then noted and the nation wide total sales are projected from these results.

Problem

Following Table gives the sales record of a firm. Determine the regression line and make a fore cast for the year 2010. Show the results graphically.

Year	Sales $\times 10^5$ (₹)
1999	45.0
2000	42.5
2001	50.1
2002	50.6
2003	62.0
2004	52.0
2005	53.5
2006	64.3
2007	60.1
2008	73.6
2009	71.0

Solution

Many methods can be used to fit a straight line to a given scatter, which suggests a linear trend, but the most widely acceptable method is that of least squares. By this method a straight line is defined in such a way that the sum of the squares of the differences between ordinates of the suggested line and those of the given points is at a minimum. The data in the above Table may be represented by coordinates, so that the first year (1999) is denoted by $x_1 = 0$, the second year by $x_2 = 1$, etc. and the appropriate sales figures by $y_1 = 45$, $y_2 = 42.5$ etc. The straight line that we want to fit in (called the regression line) is expressed by the equation

$$y = a + bx$$

where a is the intercept on Y axis (at $x = 0$) and b is the slope of the line. The values of a and b obtained by making use of the condition of the least squares are given as:

$$a = \frac{(\sum y \sum x^2) - (\sum x \sum xy)}{n \sum x^2 - (\sum x)^2} \quad \text{and} \quad b = \frac{n \sum xy - (\sum x \sum y)}{n \sum x^2 - (\sum x)^2}$$

where n is the total number of points

Year	Sales $\times 10^5$ (₹) (y)	x	x^2	xy
1999	45.0	0	0	0
2000	42.5	1	1	42.5
2001	50.1	2	4	100.2
2002	50.6	3	9	151.8
2003	62.0	4	16	248.0
2004	52.0	5	25	260.0
2005	53.5	6	36	321.0
2006	64.3	7	49	450.1
2007	60.1	8	64	480.8
2008	73.6	9	81	662.4
2009	71.0	10	100	710.0
Totals	$\sum y = 624.7$	$\sum x = 55$	$\sum x^2 = 385$	$\sum xy = 3426.8$

Results from this table can be substituted to obtain the values of a and b

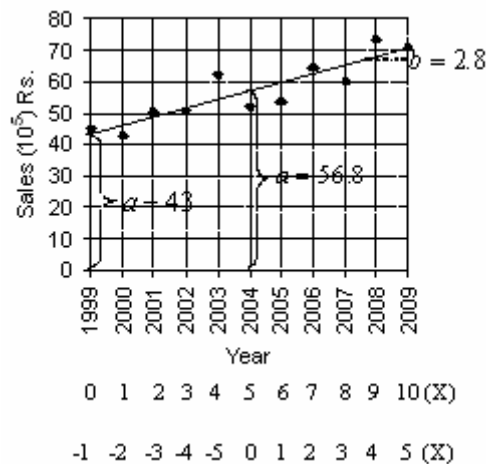
$$a = \frac{(\sum y \sum x^2) - (\sum x \sum xy)}{n \sum x^2 - (\sum x)^2} = 43.0$$

$$b = \frac{n \sum xy - (\sum x \sum y)}{n \sum x^2 - (\sum x)^2} = 2.8$$

Substituting the values of a and b , the equation of the line is:

$$y = 43.0 + 2.8x$$

The sales figures along with the trend line are plotted in the graph given below.



Sales for the year 2010 can be obtained by substituting $x=11$ in the equation $y = 43.0 + 2.8x$.

$$\Rightarrow y = 43.0 + 2.8 \times 11 = 73.80$$

That is, Sales for the year 2010 = ₹ 73.8×10^5

The computation may be somewhat simplified when x is selected in such a way that $\sum x = 0$, so that the formulae for a and b reduces to

$$a = \frac{\sum y}{n} \quad \text{and} \quad b = \frac{\sum xy}{\sum x^2}.$$

Year	Sales $\times 10^5$ (₹) (y)	x	x^2	xy
1999	45.0	-5	25	-225.0
2000	42.5	-4	16	-170.0
2001	50.1	-3	9	-150.3
2002	50.6	-2	4	-101.2
2003	62.0	-1	1	-62.0
2004	52.0	0	0	0
2005	53.5	+1	1	+53.5
2006	64.3	+2	4	+128.6
2007	60.1	+3	9	+180.3
2008	73.6	+4	16	+294.4
2009	71.0	+5	25	+355.0
Totals	$\sum y = 624.7$	$\sum x = 0$	$\sum x^2 = 110$	$\sum xy = 303.3$

$$a = \frac{\sum y}{n} = 56.8 \quad \text{and} \quad b = \frac{\sum xy}{\sum x^2} = 2.8$$

Sales for the year 2010 can be obtained by substituting $x = 6$ in the equation $y = 56.8 + 2.8x$.

Market Segmentation

The division of a market into different homogeneous groups of consumers is known as market segmentation. Rather than offer the same marketing mix to vastly different customers, market segmentation makes it possible for firms to tailor the marketing mix for specific target markets, thus better satisfying customer needs. Not all elements of the marketing mix are necessarily changed from one segment to next. For example, in some cases only the promotional campaigns will differ.

Market Segmentation can be done on the basis of the location (Geographic Segmentation); on the basis of age, income, gender and other measurable factors (Demographic Segmentation); on the basis of lifestyle, likes, dislikes, taste and preferences (Psychological Segmentation); and according to the history, loyalty and responsiveness (Behavioral Segmentation).

Human Resources Management (HRM)

Human Resources Management (or) Personnel Management is the management function that deals with recruitment, placement, training and development of organization members.

Functions of Personal Management

Functions of Human Resources Management

The functions of human resources management are listed below:

1) Procurement

- 2) Development
- 3) Compensation
- 4) Integration, and
- 5) Maintenance.

Procurement

Procurement is concerned with the obtaining of the proper quantity and quality of workforce necessary to accomplish objectives and functions of an organization. It includes the determination of human resources requirements and their recruitment and selection.

Recruitment is developing a pool of job candidates in line with the human resource plan. Candidates are usually located through newspaper and professional journal advertisements, employment agencies, word of mouth and visits to college and university campuses.

Selection is the mutual process whereby the organization decides whether or not to make a job offer and the candidate decides whether or not to accept it. The standard hiring sequence is the seven-step procedure described in the table given below. In practice, however, the actual selection process varies with different organizations and between levels in the same organization.

Steps involved in selection

PROCEDURES	PURPOSES	ACTIONS AND TRENDS
1. Completed job application	Indicates applicant's desired position; provides information for interviews	Requests only information that predicts success in the job
2. Initial screening interview	Provides a quick evaluation of applicants suitability	Asks questions on experience, salary expectation, willingness to relocate etc.
3. Testing	Measures applicants job skills and the ability to learn on the job.	May include computer testing software, handwriting analysis, mental and physical ability.
4. Background investigation	Checks truthfulness of applicants resume or application form	Calls the applicant's previous supervisor (with permission) and confirms information from applicant.
5. In-depth selection interview	Finds out more about the applicant as an individual	Conducted by the manager to whom the applicant will report.

6. Physical examination	Ensures effective performance by applicant; protects other employees against diseases, establishes health record on applicant; protects firm against unjust worker's compensation claims.	Often performed by company's medical order.
7. Job offer	Fills a job vacancy or position	Offers a salary plus benefit package.

Training and Development

The quality of employees and their development through training and education are major factors in determining long-term profitability of a concern. Both managers and non-managers may benefit from training and development programs, but the mix of experiences is likely to vary. Non – managers are often trained in the technical skills, whereas the focus of training managers will be in developing conceptual and human relations skill.

Since training involves time, effort and money it is important to be very careful while designing a training program. The objectives and need for training should be clearly identified and the method or type of training should be chosen according to the needs and objectives established. Two commonly methods of training are:

- 1) On-the-job training methods
- 2) Off-the-job training methods

On-the-job training methods

On-the-job methods enable the trainees to learn as they contribute to the aims of the enterprise. The following are some of the On-the-job training methods.

- a) Job rotation, in which the employee, over a period of time, works on a series of jobs, there by learning a broad variety of skills.
- b) Internship in which job training is combined with related class room instructions.
- c) Apprenticeship in which the employee is trained under the guidance of a highly skilled co-worker.
- d) Coaching is the training of an employee by his of her immediate superior. The superior guides his sub-ordinates and gives him or her job instructions. The superior points out the mistakes and gives suggestions for improvement.
- e) Planned work Activities, in which trainees are given work assignments to develop their experience and ability. Trainees may be asked to head a task force or participate in an important committee meeting. Such experiences help them to understand how organizations operate and also improve their human relations skill.

Off-the-job training methods

Off the job training takes place outside the workplace but attempts to simulate actual working conditions. This type of training includes:

- a) Vestibule training - Employees train on the actual equipment and in a realistic job setting, but in a room different from the one in which they will be working. The object is to avoid the on the job pressures that might interfere with the learning process.
- b) Audiovisual methods such as television, videotapes and films are employed for providing real world conditions and situations in a short time. One advantage is that the presentation is the same no matter how many times it is played. This is not true with lectures, which can change as the speaker is changed or can be influenced by outside constraints. The major flaw with the audiovisual method is that it does not allow for questions and interactions with the speaker, nor does it allow for changes in the presentation for different audiences.
- c) Computer-Assisted Instruction (CAI) can be employed which can reduce the time needed for training and also can provide more help for individual trainees. They also allow the trainee to learn at his or her own pace.
- d) Classroom Instructions- In this approach specialist from inside or outside the organization teach trainees a particular subject. Classroom instruction is often supplemented with exercises.
- e) Planned Readings, involves reading of relevant literature. The training department may aid a manager by providing a list of valuable books. This is essentially a self-development

Compensation

Compensation provides for the adequate and equitable remuneration of the work force in order to secure their best contribution to the achievement of the organisation's goals. Fixing wage rates for different categories of employees is an important task of management. The employees are not only concerned with wages received but also concerned with the level of wages received by same level of employees in similar organizations. The relative wage rules should be fixed carefully, because they have implications for promotion, transfer, seniority and other important personnel matters.

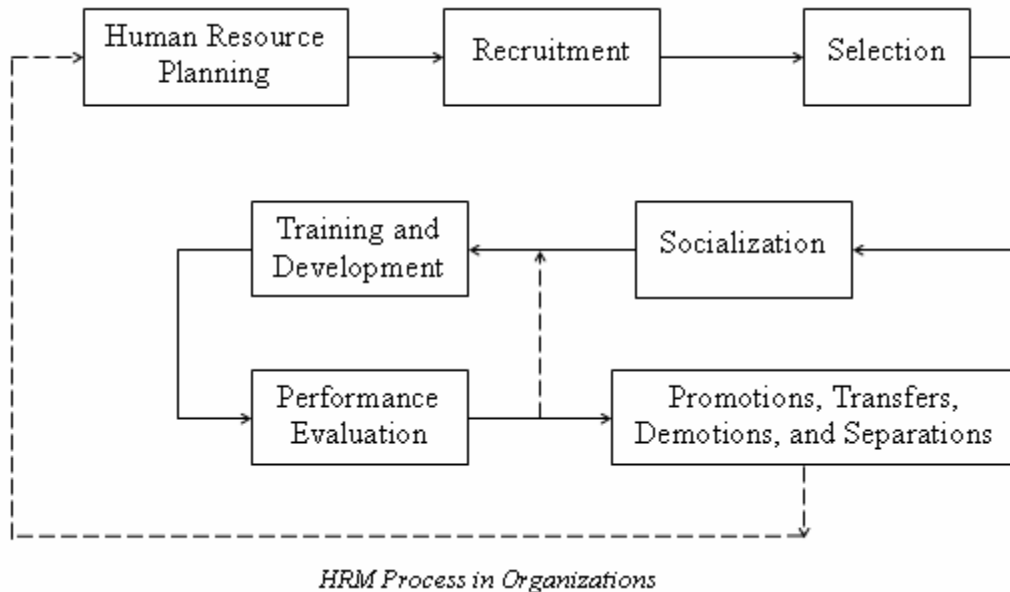
Integration

The basic objective of human resource management is to secure maximum performance from the employees. This can be accomplished only through better integration between the organisation and its employees. An effective integration between the organisation and its employees depends on three things, namely motivation, leadership and communication. In recent years the human relation exponents have revolutionized the ways and means of dealing with employees for greater performance and productivity. Hence, managerial job has become more complicated and challenging.

Maintenance

This function of personnel management is concerned with maintaining the abilities and attitudes already created and improving the conditions established through health safety, welfare and benefits programs

The above functions of the personnel management are usually represented by a figure as given below and are referred to as “HRM process” in organizations.



Wages and Incentives

Wage may be defined as payment for the use of labour and it includes money as well as non-money payments.

Wage payment plans or systems can broadly be classified into two categories:

- 1) Time rate system or non-incentive system, wherein the earnings of the worker is proportional to the time spent by him on the work. This system of wage payment is independent upon the quantity of output produced by the worker. If R is the wage rate per hour and T is the number of hours of work, then the earnings of the worker, E can be expressed as: $E = RT$
- 2) Incentive system wherein the earnings of the worker is proportional to the quantity of output produced. Some of the important wage incentive plans are described below.

Straight Piece Rate System

It is the simplest method of payment and under this method payment is made according to the number of units produced at a fixed rate per unit. For example, if a worker produces 30 units per day and for each unit the wage rate is ₹10, then the earnings of the worker will be ₹ 300 per day.

Advantages

- a) Efficiency is recognized
- b) Motivates the workers
- c) Computation of labour cost is easy
- d) Less Idle Time
- e) Helps to boost production

Disadvantages

- a) Quality suffers as the emphasis is on quantity
- b) More scrap and defective Work
- c) Damage to Tools and Equipment
- d) No Guaranteed Wages
- e) Uncertainty in Income
- f) Trade Unions Oppose the System
- g) Enforced idleness because of power failure, material shortage, etc. can badly affect the earnings of a worker

Straight Piece Rate System with a Guaranteed Wage

Under this system, a standard output is set by the management and a corresponding minimum guaranteed wage is also fixed. If a worker produces less than (or equal to) the standard output he will be getting the minimum guaranteed wage. On the other hand, if a worker exceeds this standard, he will be paid in accordance with the straight piece rate system. For example, if the output standard is 25 pieces per day, minimum guaranteed wage is ₹ 100 per day, and wage rate per piece is ₹ 4, then as per this system, a worker producing 20 units per day will get ₹ 100, and a worker producing 30 units per day will get ₹ 120.

Advantages

- a) Provides a guaranteed minimum wage
- b) The minimum wage takes care of the enforced idleness beyond the control of workers

Disadvantages

With respect to the above example, the producing 20 pieces in a day will get Rs.100 for that day @ $\text{₹ } \frac{100}{20} = \text{₹ } 5$ per piece, where as the producing 30 pieces in a day will get ₹ 120 for that day @ $\text{₹ } \frac{120}{30} = \text{₹ } 4$ per piece. This shows that the system does not offer sufficient incentive for a worker who exceeds the set output standard.

Differential Piece Rate System

In this system, up to a certain production level, which is the standard output, a piece rate is given. A worker who exceeds this output will be paid at a higher piece rate. However, this system of wage payment does not guarantee minimum base wage.

Advantages

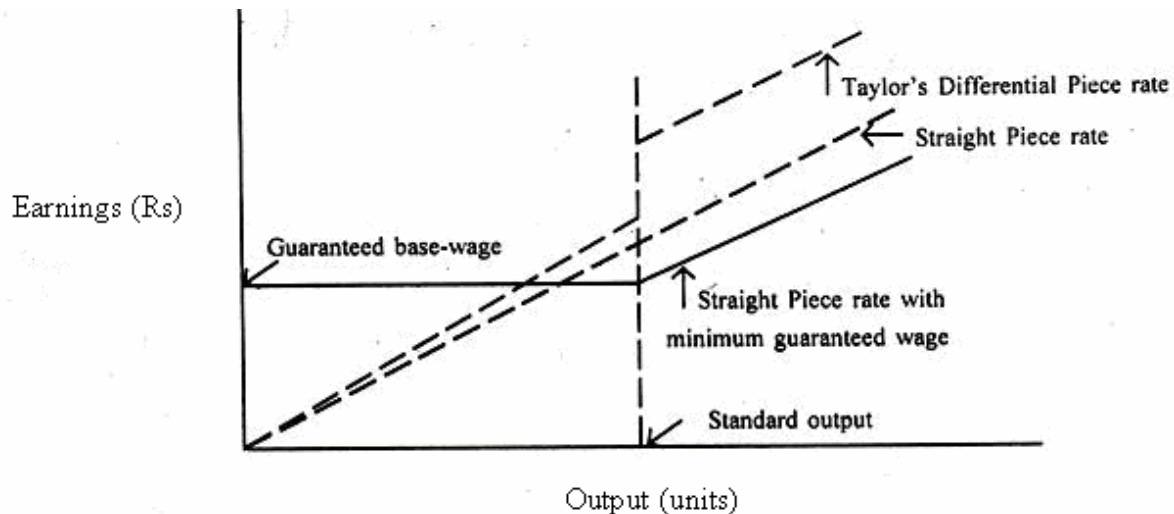
- a) Provides incentives to efficient worker

- b) Penalizes inefficient worker
- c) Focuses on high production rate
- d) Simple and easy to implement

Disadvantages

- a) Minimum wage is not assured.
- b) No consideration for the machine failure, power failure, etc.
- c) Over emphasis on high production rate
- d) There are chances that quality of work may suffer

A comparison of the three piece rate systems are given in the following figure.



Piece rate systems

Halsey Plan

Under this plan a standard time is set for the completion of a job. Workers who complete the job in less than the standard time will earn a bonus, whereas a worker who takes longer than the standard time (or even the standard time) is paid at a guaranteed time rate and will not earn any bonus.

If R is the hourly wage rate, T is the time taken by the worker to complete the job, S is the standard time allowed for the job, and p is the percentage of bonus (on saved time) given to the worker, then the earnings of the worker, E can be expressed as:

$$E = RT + \frac{p}{100}(S - T)R$$

If the second term $\left\{ \frac{p}{100}(S - T)R \right\}$ in the above equation works out to be negative, the same should be discarded.

For a 50 – 50 Halsey plan, 50% of the bonus on the time saved is enjoyed by the worker and the remaining 50% by the management, and the above equation can be simplified as:

$$E = RT + (S - T) \frac{R}{2}$$

Advantages

- a) Beneficial to efficient worker
- b) Management also shares the gain of workers
- c) Minimum base-wage is guaranteed.
- d) Simplicity of Calculation

Disadvantages

- a) Workers get only a percentage of return on their over-achievement
- b) Standard completion time based on past production data need not be fair and just to all workers

Problem

If the allowed time for a job is 10 hours and the hourly rate is ₹ 4, calculate the earnings of workers A, B, and C who completes the job in 10 hours, 12 hours, and 8 hours respectively.

Solution

Worker	Earnings the Worker
A	$E = RT + (S - T) \frac{R}{2} = 4 \times 10 + 0 = ₹ 40$ (for 10 hours)
B	$E = RT + (S - T) \frac{R}{2} = 4 \times 12 = ₹ 48$ (for 12 hours)
C	$E = RT + (S - T) \frac{R}{2} = 4 \times 8 + (10 - 8) \frac{4}{2} = ₹ 36$ (for 8 hours)

If the workers A, B, and, C are allowed to work for a fixed time (say, 8 hours), then they will earn ₹ 32, ₹ 32, and ₹ 36 respectively.

Rowan Plan

Rowan plan is similar to Halsey plan except the way in which the bonus is calculated. If R is the hourly wage rate, T is the time taken by the worker to complete the job, and S is the standard time allowed for the job, then the earnings of the worker, E can be expressed as:

$$E = RT + \frac{S - T}{S} RT$$

Advantages

- a) Minimum base-wage is guaranteed
- b) Checks over-speeding because once the time saved increases beyond a particular limit the bonus will tend to decrease
- c) Management also shares the gain of workers

Disadvantages

- a) Insufficient incentives for highly efficient workers

- b) Difficult in ascertaining wages as it requires large data processing
 c) Discourages overachievers

Problem

The standard time and rate for the subassembly of a television are fixed as 6 hours and ₹ 50 per hour respectively. Calculate the bonus, total earnings, and effective hourly rate of four workers A, B, C, and D, if they save, 1 hour, 2 hours, 3 hours, and 4 hours respectively from the standard time under Rowan plan.

Solution

$S = 6$ hours

$R = ₹ 50$ per hour

The total earnings (or wage) as per the Rowan plan can be expressed by the equation:

$$E = RT + \frac{S-T}{S} RT$$

In the above equation, the first term on the right hand side, (RT) represents the rated income and the second term $\left\{ \frac{S-T}{S} RT \right\}$ represents the bonus or incentive. Effective hourly rate can be obtained by dividing the total earnings with the actual time taken by the worker. The results are tabulated in the following table.

Calculation of bonus, total earnings, and effective hourly rate

(1) Worker	(2) Time Taken = R (hours)	(3) Rated Income = RT (₹ per hour)	(4) Incentive = $\left\{ \frac{S-T}{S} \right\} RT$ (₹)	(5) = (3)+(4) Total Wages (₹)	(6) = (5) ÷ (2) Effective Hourly Rate (₹ per hour)
A	5	250	41.67	291.67	58.33
B	4	200	66.67	266.67	66.67
C	3	150	75	225	75
D	2	100	66.67	166.67	83.33

It can be seen from the above tabulation that the incentive for Rowan plan will be maximum when $T = \frac{S}{2}$

Emerson Efficiency Plan

This plan guarantees wages on time rate. In addition to this guaranteed time rate, the workers who prove efficient are paid a bonus. For the purpose of determining efficiency, the standard completion time for the job is determined and the following equation is employed:

$$\text{Efficiency of a worker, } \eta = \frac{\text{Standard completion time of the job}}{\text{Actual time taken by the worker to complete the job}}$$

The bonus payable at different efficiency levels are given in the following table.

Efficiency, η	Bonus
Less than 66 $\frac{2}{3}$ %	No bonus is payable
66 $\frac{2}{3}$ %	1% on wages earned as per time rate
90%	10% on wages earned as per time rate
100%	20% on wages earned as per time rate
Above 100%	30% on wages earned as per time rate

Advantages

- a) Guarantees minimum wage
- b) Efficient worker is rewarded handsomely
- c) Applicable to group of workers

Disadvantages

- a) Disproportionate rate of bonus below standard output
- b) Chances of over-speeding and compromise of quality
- c) Involves more paper work

Characteristics of an Ideal Incentive Plan

- 1) The plan should be simple to understand and administer
- 2) The plan should guarantee a minimum base wage
- 3) The plan should ensure that the quality of production is maintained
- 4) Standards should be scientifically set and comparable with other industries of similar type
- 5) The plan should be fair for both employee and employer
- 6) The plan should encourage workers to perform more and better
- 7) The plan should ensure that resources are utilized properly and there is no undue wastage
- 8) The plan should be implemented only after thorough discussions and mutual consent of the employee and employer
- 9) The plan once installed should be maintained properly and should not be subjected to frequent changes

Financial management

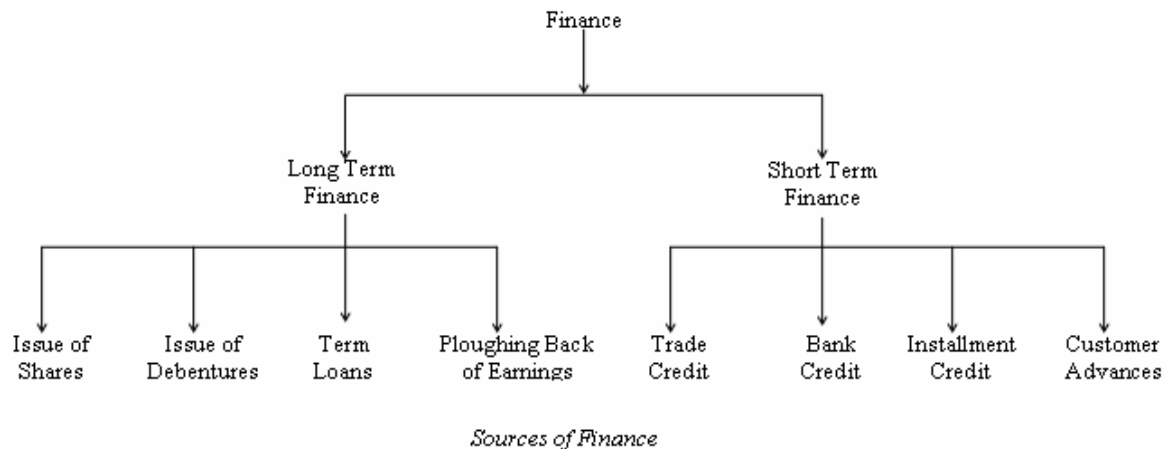
Financial management is related with the acquisition and use of funds by an organization.

The major tasks (or functions) of financial management are as listed below:

- 1) Financial planning and control
- 2) Determination of capital requirements
- 3) Acquisition of funds by determining the financing mix
- 4) Allocation of capital funds
- 5) Management of current assets
- 6) Determination of dividend payout ratio

Sources of Finance

One of the major functions of financial management is determining the financing mix for raising the required amount of capital to run the business. This requires a clear understanding of the available sources of money. Depending upon the period for which the funds are required the sources of finance can be classified as given below.



Issue of Shares

The most important source of long term fund is issue of shares. The company invites members of the public through its prospectus to purchase shares and the persons who purchase the shares are called share holders. The liability of the share holders is limited to their capital contributions and they are entitled to transfer shares. As per the Indian Companies Act, 1956 a company can issue two classes of shares namely *preference shares* and *equity (or) ordinary shares*. Preference shares are those which carry a preference with regard to payment of dividend and repayment of capital in the event of winding up. The rate of dividend on a preference share is fixed and must be paid before any dividend is paid on other classes of shares. Ordinary shares will not have any preferences and dividend on these shares is paid only after payment of dividend to preference share holders. The rate of dividend fixed on equity shares is not fixed and it may vary from year to year depending on the financial position of the company.

Issue of Debentures

A debenture is an acknowledgement of debt issued under the seal of the company. Debentures are issued by the company when it wants to raise the money without increasing its share capital.

There are many differences between a debenture holder and a share holder. A share holder is a part owner of the company, but a debenture holder is only a creditor. A share holder does not receive any dividend unless the company makes a profit, but the debenture holder will receive the interest on money lent by him, whether or not the company has made a profit. In the event of winding up of the company the amount to debenture holders must be paid back before any amount is paid to share holders.

Term Loans

Term loan refers to long term financial assistance that is generally repayable in more than one year but less than ten years. They represent secured borrowings. The repayment of principal amount of the term loan plus interest is a statutory obligation irrespective of the financial position of the company.

The following factors are generally considered by the lending institution before granting a term loan.

- 1) Type of company
- 2) Type of technology to be adopted by the company
- 3) Location of the company
- 4) Goods and services required by the company and its availability
- 5) Projected demand of the output of the product concerned and the expected revenue
- 6) Managerial competence
- 7) Repayment capacity

Ploughing Back of Earnings

Under this system, the entire profit generated by the company will not be distributed among the share holders or owners, but a portion of it will be retained by the company to serve as a source of finance for expansion and growth of the concern. This is also referred to as *internal financing or self financing*.

Trade Credit

This represents the credit granted by other firms or suppliers, usually for a period of thirty to ninety days. Trade credit is usually granted to the company on the basis of its good will and also the financial position.

Bank Credit

This is an important source of short term finance and may be in the form of advances, loans, cash credits, overdrafts, etc.

Installment Credit

This type of finance is generally employed by the companies to purchase equipments and machinery and the payment is made on easy installments.

Customer Advances

Short-term financial requirements of the company, at least partly, can be met through customer advances. These advances are obtained from the customers at the time of ordering or booking of the product and it represents a part of the product cost. The period of such credit will depend upon the time taken to deliver the product.

Bond Value

A bond or a debenture is essentially a long-term note given to the lender by the business, stipulating the terms of repayment and other conditions.

Bonds usually are issued in units ranging ₹100 to ₹1000, which is known as the *face value or par value* of the bond. This represents the amount to be repaid at the time of maturity. The interest rate quoted on the bond is called *coupon rate or bond rate*, and the periodic interest payable to the bond holder is simply the face value times the coupon rate per period.

The value of a bond, at any time, is the present worth of future cash receipts associated with it and is expressed by the equation:

$$V = \frac{Z}{(1+i)^n} + rZ \frac{(1+i)^n - 1}{i(1+i)^n}$$

where,

V = Value of the bond

Z = Face value or par value of the bond

r = Coupon rate or bond rate per interest period

i = Rate of return or bond yield rate per period

n = Maturity period of the bond

The quantity $\frac{1}{(1+i)^n}$ in the above equation is called the single payment present worth

factor and is usually denoted by the symbol $(P/F, i, n)$. Similarly the quantity $\frac{(1+i)^n - 1}{i(1+i)^n}$

is called the uniform series present worth factor and is usually denoted by the symbol $(P/A, i, n)$. Hence, the value of the bond can also be expressed as:

$$V = Z(P/F, i, n) + rZ(P/A, i, n)$$

Problem

A bond with a face value of ₹1000 is bearing a coupon rate of 8% and interest is payable annually. This bond will mature after 20 years, and the first interest payment is due one year from now.

- How much should be paid now for this bond in order to receive a yield of 10% per year on the investment?
- If this bond is purchased now for ₹900, what yield would the buyer receive?

Answer

a) Value of the bond can be expressed as $V = \frac{Z}{(1+i)^n} + rZ \frac{(1+i)^n - 1}{i(1+i)^n}$

$$Z = ₹1000$$

$$rZ = (0.08)1000 = ₹80$$

$$i = 0.1$$

$$n = 20 \text{ years}$$

$$V = \frac{Z}{(1+i)^n} + rZ \frac{(1+i)^n - 1}{i(1+i)^n}$$

Lecture Note on Industrial Organization and Management

Subject handled by: Dr. Shouri P.V., Associate Professor in Mechanical Engineering, MEC, Cochin
(for S₇ Computer Science Batches)

$$V = \frac{1000}{(1+0.1)^{20}} + 80 \frac{(1+0.1)^{20} - 1}{0.1(1+0.1)^{20}} = ₹829.69$$

b) The value of the bond is given as $V = ₹900$ and the yield that a buyer will get, i , has to be evaluated.

$$\text{That is, } 900 = \frac{1000}{(1+i)^{20}} + 80 \frac{(1+i)^{20} - 1}{i(1+i)^{20}}$$

The value of i has to be obtained from the above equality by trial and error. $i = 9.1$ will be a good approximation to satisfy the above equality.

An approximate value of i can also be obtained using the following formula:

$$i = \frac{rz + (Z - V)/n}{0.4Z + 0.6V}$$

Problem

A ₹100 par value bond bearing a coupon rate of 12% will mature after 5 years. What is the value of the bond, if the discount rate is 15%.

Answer

$$V = \frac{Z}{(1+i)^n} + rZ \frac{(1+i)^n - 1}{i(1+i)^n} = \frac{100}{(1+0.15)^5} + (0.12)(100) \frac{(1+0.15)^5 - 1}{0.15(1+0.15)^5} = ₹89.9464$$

Problem

The market price of a ₹1000 par value bond carrying coupon rate of 14% and maturing after 5 years is ₹1050. What is the yield to maturity on this bond?

Answer

$$V = \frac{Z}{(1+i)^n} + rZ \frac{(1+i)^n - 1}{i(1+i)^n}$$

$$1050 = \frac{1000}{(1+i)^5} + (0.14)(1000) \frac{(1+i)^5 - 1}{i(1+i)^5}$$

$i = 12.6$ will satisfy the above equality.

Valuation of Equity Shares

A very simple model for the valuation of equity share is the dividend valuation model. A share holder in company is entitled to receive cash dividends declared by the company as well as the price of the share at the time it is sold. The current price of an equity share in the dividend valuation model can be approximated by the present worth of future cash receipts during an n -year ownership period and can be expressed as:

$$P_0 = \frac{D_1}{(1+i)} + \frac{D_2}{(1+i)^2} + \dots + \frac{D_n}{(1+i)^n} + \frac{P_n}{(1+i)^n}$$

where i = Rate of return per year required by the share holders

P_0 = Current price of the equity share

P_n = Selling price of the equity share at the end of n years

$$\begin{aligned}
 D_1 &= \text{Dividend expected for the 1}^{\text{st}} \text{ year} \\
 D_2 &= \text{Dividend expected for the 2}^{\text{nd}} \text{ year} \\
 D_n &= \text{Dividend expected for the } n^{\text{th}} \text{ year}
 \end{aligned}$$

If it is assumed that the dividends are constant at a value of D for an infinitely long duration, then the current price of equity share equals the present worth of an infinite series of dividend receipts that remain constant in amount can be expressed as:

$$P_0 = D(P/A, i, \infty) = D \frac{(1+i)^\infty - 1}{i(1+i)^\infty}$$

Dividing the numerator and denominator by $(1+i)^\infty$, the equation for P_0 becomes:

$$P_0 = D \left\{ \frac{1 - \frac{1}{(1+i)^\infty}}{i} \right\} \Rightarrow P_0 = D \left(\frac{1-0}{i} \right) = \frac{D}{i}$$

Instead of assuming a constant dividend stream if the dividends are assumed to grow at constant compound rate of g , then the equation for the current price of the equity share

$$\text{can be expressed as } P_0 = \frac{D}{i - g}$$

As ' D ' in the above equation is the dividend corresponding to year 1, it will be more appropriate to write the above equation as:

$$P_0 = \frac{D_1}{i - g}$$

$$\begin{aligned}
 \text{where } D_1 &= \text{Dividend expected for the 1}^{\text{st}} \text{ year} \\
 g &= \text{Constant compound growth rate} \\
 i &= \text{Expected rate of return}
 \end{aligned}$$

Problem

A public limited company's dividends per share for the next two years are expected to be ₹ 3 and ₹4 respectively. There after the dividend is expected to grow at a constant rate of 6% forever. If the investor's rate of return on the company's equity is 15%, calculate the share value.

Answer

The expected dividends at the end of year 1 and year 2 are $D_1 = ₹3$ and $D_2 = ₹4$ respectively. From 3rd year onwards the dividend is expected to grow at a constant rate of 6% forever. So dividend for the 3rd year will be $D_3 = 4 + 4 \times \frac{6}{100} = ₹4(1.06)$ and the equivalent of infinite cash flows growing at a constant rate of 6% thereafter can be brought to year 2 by making use of the equation $P_2 = \frac{D_3}{i - g} = \frac{4(1.06)}{(0.15 - .06)} = ₹47.11$

$$\text{Thus the present value of the share is } P_0 = \frac{D_1}{(1+i)} + \frac{D_2}{(1+i)^2} + \frac{P_2}{(1+i)^2}$$

$$\Rightarrow P_0 = \frac{3}{(1+0.15)} + \frac{4}{(1+0.15)^2} + \frac{47.11}{(1+0.15)^2} = ₹41.26$$

Book-keeping and Accounting

Book-keeping is the making of routine records, individually and in groups, of financial transactions according to a set of rules.

Accounting is the summarization of the recorded information and analyzing and interpreting it for the use of the concerned parties to aid them in decision making.

Book-keeping is only a preliminary aspect of accounting.

Systems of Accounting

Two systems of accounting are in common use. They are:

- 1) The single entry system
- 2) The double entry system

Every transaction is between two persons or two concerns. Every transaction, thus, affects two accounts. Single entry system records only one side of the transaction and hence it does not provide complete information about a transaction. This system of book-keeping is not generally used.

Double entry system records both the sides of the transaction and thus provides complete information of the business transaction. That is, it is based on the dual aspect concept. The double entry system of accounting is more scientific and is compulsory for joint stock companies in India.

Accounting concepts

- 1) Business entity concept - business is separate entity from owner.
- 2) Dual Aspect concept - Liabilities = Assets (*Dr. = Cr.*)
- 3) Going concern concept - business is going to be in existence for an indefinitely long time.
- 4) Accounting period concept - Indefinite long period is divided into short span for accounting purpose.
- 5) Cost concept - cost of acquisition of assets is considered for accounting (considering depreciation) and not current price of assets.
- 6) Money measurement concept - only facts which can be measured in money find place in accounting.
- 7) Matching concept - expenses and costs incurred during period whether paid or not must match the revenue for that particular period.

The Journal

The Journal is a chronological record of business transactions. The first record of a transaction is made in this book. It is book of first or original entry in which transactions are recorded one after another, in the order in which they occur, showing:

- (1) date for each transaction
- (2) account and amount to be debited
- (3) account and amount to be credited, and
- (4) an explanation (if necessary).

The function of a journal is to provide a permanent and complete record, arranged in chronological order for future reference, of all business transactions of a firm. Every page of a journal is numbered for future reference. The usual form of a journal is given below.

JOURNAL

Page No:

Date	Particulars	Ledger Folio (L.F.)	Debit Amount (₹)	Credit Amount (₹)

Note: The Ledger folio (L.F.) column in the Journal is intended to write the page (folio) number of the ledger where the particular account is opened.

Types of Accounts

Accounts can be of three types, namely:

1. Real Accounts
2. Personal Accounts
3. Nominal Accounts

Real accounts are accounts of assets or properties.

Personal accounts are accounts in the names of persons, firms or companies.

Nominal accounts related with elements such as wages, salaries, rent, commission, and interest.

Rules for Debit and Credit

1. In the case of real accounts, debit what comes in and credit what goes out.
2. In the case of personal accounts, debit the receiver and credit the giver.
3. In the case of nominal accounts debit expenses or losses and credit gains or profits.

Ledger

A group of accounts is known as a *ledger*. After the transactions have been recorded in the Journal, the accounts are prepared in this book. Ledger contains the same information (as given in journal) but properly arranged according to each person or firm it is related to. The ledger, therefore, is a derived or secondary record. The usual form of an account is given below.

The account contains two identical halves. The left side of an account is known as the debit side and is for debit entries. The right side of an account is known as the credit side and is for credit entries. The process of transferring journal entries to the ledger is known as posting.

LEDGER

Dr.				Page Number:		Cr.	
Date	Particulars	Journal Folio (JF)	Amount (₹)	Date	Particulars	Journal Folio (JF)	Amount (₹)
	To. " " (Debit side)				By. " " (Credit side)		

Problem

Journalize and post the following transactions:

2010

- January 1 - Mr. X started business by investing an amount of ₹ 5000.
- January 8 - Purchased goods worth ₹1000.
- January 10 - Paid ₹ 50 as commission.
- January 15 - Sold grinding machine for ₹ 500.
- January 16 - Paid ₹ 2000 to bank.
- January 21 - Received ₹ 100 as commission from Mr. B
- January 31 - Paid ₹ 200 as office rent

JOURNAL

Page Number: 1

Date	Particulars	Ledger Folio (LF)	Dr. (₹)	Cr. (₹)
2010, January 1	Cash A/c Dr. To Mr. X, Capital A/c	1 2	5000	5000
January 8	Goods A/c Dr. To Cash A/c	3 1	1000	1000
January 10	Commission A/c Dr. To Cash A/c	4 1		
January 15	Cash A/c Dr. To Goods A/c	1 3	500	500
January 16	Bank A/c Dr. To Cash A/c	5 1	2000	2000
January 21	Mr. B. Dr. To Commission A/c	6 4	100	100
January 31	Rent A/c Dr. To Cash A/c	7 1	200	200
<i>Total</i>			8850	8850

Ledger – Cash Account

Dr.

Cr.

Page No:1

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
2010 Jan 1	To Mr. X, Capital	1	5000	2010 Jan 8	By Goods	1	1000
Jan 15	" Goods	1	500	Jan10	" Commission	1	50
				Jan16	" Bank	1	2000
				Jan31	" Rent	1	200
					" Balance		2250
			5500				5500

Note: The difference between the two sides of an account is known as an account balance. If the debit entries in an account exceed the credit entries, the account has a debit balance. If the credits exceed the debits, the account has a credit balance. For e.g., in the above cash account, the total of credits equals $1000 + 50 + 2000 + 200 = 3250$, and the total of debits equals $5000 + 500 = 5500$. This means the account has a debit balance of $5500 - 3250 = 2250$ and the debit balance is entered on the credit side of the account as indicated above so that the two sides are balanced. Similarly, if an account has a credit balance it will be entered on the debit side.

Ledger – Mr. X, Capital Account

Dr.

Cr.

Page No:2

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
	To Balance		5000	2010 Jan 1	By Cash	1	5000
			5000				5000

Ledger – Goods Account

Dr.

Cr.

Page No:3

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
2010 Jan 8	To cash	1	1000	2010 Jan15	By cash	1	500
					" Balance		500
			1000				1000

Ledger – Commission Account

Dr.

Cr.

Page No:4

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
2010 Jan10	To cash	1	50	2010 Jan21	By Mr. B	1	100
	" Balance		50				
			100				100

Ledger – Bank Account

Dr.

Cr.

Page No:5

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
2010 Jan16	To cash	1	2000	2010 Jan21	By Balance		2000
			2000				2000

Ledger – Mr. B

Dr.

Cr.

Page No:6

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
2010 Jan21	To commission	1	100		By Balance		100
			100				100

Ledger – Rent Account

Dr.

Cr.

Page No:7

Date	Particulars	Journal Folio (J. F.)	Amount (₹)	Date	Particulars	Journal Folio (J. F.)	Amount (₹)
2010 Jan 31	To Cash	1	200		By Balance		200
			200				200

Trial Balance

If no book-keeping errors are made the total of the debit balances and credit balances in the ledger will be equal. A list of ledger account balances prepared for proving the equality of debits and credits in the ledger on a certain date is called a *Trial Balance*. The trial balance of the above transactions will be as follows:

Trial Balance, January 31, 2010

Account	Debit balances (₹)	Credit balances (₹)
Cash	2250	-
Capital	-	5000
Goods	500	-
Commission	-	50
Bank	2000	-
Mr. B	100	-
Rent	200	-
Total	5050	5050

Cash Book

The cash book is the most important subsidiary book in any business concern. It is used to record cash transactions. When cash is received cash account is debited and when cash is paid cash account is credited in the journal. The other account involved is either debited or credited as the case may be. But it is not necessary to generalize each cash transactions when a cash book is maintained. All cash transactions are first entered in the cash book. The cash book has two sides, a receipt side or debit side and a payment side or credit side. All receipts of cash are entered on the debit side, while all payments of cash are entered on the credit side. The usual form of a simple cash book is given below.

CASH BOOK

Dr.					Cr.				
Date	Particulars	Receipt No.	L.F.	Amount (₹)	Date	Particulars	Voucher No.	L.F.	Amount (₹)
	(Receipts)					(Payments)			

It can be seen that the cash book is simply a cash account. When a cash book is maintained, a separate cash account need not be opened in the ledger. The cash book may be balanced just as any other ledger account. The balance represents '*cash in hand*'

which should tally with the actual cash with the cashier. Thus the cash book serves two purposes. It is a subsidiary book, a book of original entry. It is also a principal book or a book of final entry as it serves the purpose of a ledger account as regards cash.

Financial Statements

Financial statement refers to the monetary analysis of the flow of goods and services to, within, and from the organization. There are two basic financial statements connected with any organization, namely: (a) balance sheet and (b) income statement or trading and profit and loss account.

Balance sheet

The balance sheet describes the company in terms of its assets, liabilities, and net worth as at the date of its preparation and the accounting equation defining the balance sheet is :

$$\text{Assets} - \text{Liabilities} = \text{Net worth}$$

Assets are things owned by the company and it may range from money in hand to the good will value of its name in the market place. The two classes of assets distinguishable are *current assets* and *fixed assets*. Current assets represent the short-lived working capital of the company that can be converted into cash approximately within a year. Fixed assets generally have long life and if it is to be converted into cash there must be some major change in the business.

Liabilities are the obligations to the company and they are also made up of two groups, namely *current liabilities* and *fixed liabilities*. Current liabilities are generally payable in a year and only long terms debts that are due in ten years or more can be considered as fixed liabilities.

The company's net worth is the residual value remaining after total liabilities have been subtracted from total assets.

Income statement

While the balance sheet describes a company's financial condition at a given *point in time*, the income statement summarizes the company's performance over a given *interval of time* and is defined by the accounting equation:

$$\text{Revenue} - \text{Expenses} = \text{Profit (or Loss)}$$

The income statement consists of two parts, namely: (1) the trading account, which shows the gross profit (or loss), and (2) the profit and loss account that shows the net profit (or loss).

In a trading account, opening stock, net purchases (i.e., purchases minus returns outward), and direct expenses (i.e., wages and carriage inwards) are entered on the debit side, and the credit side consists of net sales (i.e., sales minus returns inward), and the closing stock. A credit balance will imply a gross profit for the company and a debit balance will imply a gross loss.

Profit and loss account is usually prepared as a continuation of trading account, and all other expenses and revenues that are associated with the business and are not shown in the trading account will be taken care of by this account. A credit balance will imply a net profit for the company and a debit balance will imply net loss.

Problem

From the following Trial Balance prepare Trading and Profit and Loss Account for the year ended 31.12.2009 and Balance Sheet as on 31.12.2009.

Dr.		Cr.	
	(₹)		(₹)
Drawings	10000	Capital	81000
Plant	60000	Sundry creditors	45000
Sundry debtors	40000	Sales	140000
Purchases	80000	Purchase returns	4500
Sales returns	4000		
Wages	15000		
Cash in Hand	1000		
Cash at bank	6000		
Salaries	11000		
Repairs	4000		
Rent	4500		
Opening stock	20000		
Bills receivable	15000		
Total	270500		270500

The closing stock was valued at ₹ 28,000.

Solution

TRADING AND PROFIT AND LOSS ACCOUNT for the year ended 31-12-2009

Dr.		Cr.	
	(₹)		(₹)
To Opening stock	20000	By Sales	140000
" " Purchases 80000		Less returns 4000	136000
" " Less returns 4500	75500	" " Closing stock	28000
" " Wages	15000		
" " Gross profit (c/d)	53500		
Total	164000		164000
To Salaries	11000	By Gross profit (b/d)	53500
" " Repairs	4000		
" " Rent	4500		
" " Net profit	34000		
Total	53500		53500

BALANCE SHEET as on 31-12-2009

LIABILITIES	(₹)	ASSETS	(₹)
Sundry creditors	45000	Cash in hand	1000
		Cash at bank	6000
		Bills receivable	15000
NET WORTH		Sundry debtors	40000
Capital : 81000		Closing stock	28000
Less drawings : <u>10000</u>	71000	Plant	60000
Net Profit (added)	34000		
Total	150000		150000

Problem

From the following Trial Balance prepare Trading and Profit and Loss Account for the year ended 31.12.2009 and Balance Sheet as on 31.12.2009.

Dr.		Cr.	
	(₹)		(₹)
Drawings	10000	Capital	81000
Plant	60000	Sundry creditors	45000
Sundry debtors	40000	Sales	140000
Purchases	80000	Purchase returns	4500
Sales returns	4000		
Wages	15000		
Cash in Hand	1000		
Cash at bank	6000		
Salaries	11000		
Repairs	4000		
Rent	4500		
Opening stock	20000		
Bills receivable	15000		
Total	270500		270500

The following adjustments are to be made:

1. Wages outstanding ₹ 600
2. Rent paid in advance ₹ 500
3. Depreciate plant by 10 % p.a.
4. Closing stock was valued at ₹ 28000

SolutionTRADING AND PROFIT AND LOSS ACCOUNT for the year ended 31-12-2009

Dr.	(₹)		Cr.
To Opening stock	20000	By sales 140000	
" " Purchases 80000		Less returns <u>4000</u>	136000
Less returns <u>4500</u>	75500	" " Closing stock	28000
" " Wages 15000			
Add o/s wages <u>600</u>	15600		
" " Gross profit (c/d)	52900		
Total	164000		164000
To Salaries	11000	By Gross profit (b/d)	52900
" " Repairs	4000		
" " Rent 4500			
Less advance <u>500</u>	4000		
" " Depreciation	6000		
" " Net profit	27900		
Total	52900		52900

BALANCE SHEET as on 31-12-2009

LIABILITIES	(₹)	ASSETS	(₹)
Sundry creditors	45000	Cash in hand	1000
Wages outstanding	600	Cash at bank	6000
		Bills receivable	15000
		Rent advance	500
NET WORTH		Sundry debtors	40000
Capital : 81000		Closing stock	28000
Less drawings : 10000	71000	Plant 60000	
Net Profit (added)	27900	Less depreciation <u>6000</u>	54000
Total	144500		144500

Problem

The following Trial Balance was extracted from the books of a trader on Dec. 31, 2009. You are requested to prepare his Trading and Profit and loss account for the year ended on that date and the balance sheet as on that date.

	Dr. (₹)	Cr. (₹)
Capital		35000
Purchases and Sales	44500	65000
Stock (Jan 1, 2009)	26000	
Returns Outwards		600
Returns Inwards	1000	
Salaries	4750	
Trade expenses	2050	
Carriage	500	
Bad debts	230	
Discount account (balance)		350
Sundry debtors	35750	
Sundry creditors		19600
Insurance	220	
Fixtures and Fittings	1850	
Motor Vans	2500	
Rent, Rates and Taxes	3550	
Bank Overdraft		3950
Drawings	1450	
Cash in hand	150	
Total	124500	124500

The value of stock on hand on Dec 31, 2009 was Rs. 17,950.

Solution:**TRADING AND PROFIT AND LOSS ACCOUNT for the year ended 31-12-2009**

Dr.	(₹)		Cr.	(₹)
To opening stock	26000	By sales	65000	
" " Purchases 44500		Less returns 1000		64000
Less returns 600	43900	" " Closing stock		17950
" " Carriage 500				
" " Gross profit (c/d)	11550			
Total	81950			81950
To salaries	4750	By Gross profit (b/d)	11550	
" " Trade expenses 2050		" " Discount A/c (balance)	350	
" " Bad debts 230				
" " Insurance 220				
" " Rent, Rates and Taxes 3550				
" " Net profit 1100				
Total	11900			11900

BALANCE SHEET as on 31-12-2009

LIABILITIES	(₹)	ASSETS	(₹)
Sundry creditors	19600	Cash in hand	150
Bank overdraft	3950	Sundry debtors	35750
		Closing stock	17950
		Fixtures and fittings	1850
		Motor vans	2500
NET WORTH			
Capital : 35000			
Less drawings : 1450	34650		
Net Profit (added)	1100		
Total	58200		58200

Valuation of Stock

During product manufacturing, at different intervals of time, the materials that are needed for production will be withdrawn from the stores. As these materials are made from purchases over a period of time and at different prices, it is not always easy to arrive at the actual cost of material being used. The accounting method that a company adopts for valuing materials used on products can directly have an impact on the final accounts of a company. The three commonly used methods of stock valuation are listed below.

1) First-In, First-Out (FIFO)

This method assumes that the material that is purchased earliest is the first used. For example, consider that at time T_1 200 kilograms of material is bought at a price of ₹ 50/kg. and 200 kilograms more at time T_2 at ₹ 60/kg. FIFO states that if 200 kilograms of material is withdrawn at time T_3 , then the withdrawal from stock is valued at ₹ $50 \times 200 = ₹ 10000$

2) Last-In, First-Out (LIFO)

This method assumes that the material that is purchased latest is the first used. 200 kilograms of material that is withdrawn at T_3 will be valued at ₹ $60 \times 200 = ₹ 12000$

3) Weighted Average

This method takes the weighted average of all the materials available at the store at the time of issue and then uses the average cost to determine the value of stock. In the present example, the average cost of material would be calculated as $\frac{200 \times 50 + 200 \times 60}{400} = ₹ 55/kg$ Using this average cost, 200

kilograms that is withdrawn at T_3 will be valued at ₹ $55 \times 200 = ₹ 11000$

Inflation

Inflation refers to the general increase in the prices of goods and services. It is measured as an annual percentage increase. As inflation rises, every Rupee you own buys a smaller percentage of a good or service. The various types of inflation are briefed below.

Demand-pull Inflation

This type of inflation occurs when total demand for goods and services in an economy exceeds the supply of the same. When the supply is less, the prices of these goods and services would rise, leading to a situation called as demand-pull inflation. This type of inflation affects the market economy adversely during the wartime.

Cost-push Inflation

As the name suggests, if there is increase in the cost of production of goods and services, there is likely to be a forceful increase in the prices of finished goods and services. For instance, a rise in the wages of workers would raise the unit costs of production and this would lead to rise in prices for the related end product. This type of inflation may or may not occur in conjunction with demand-pull inflation.

Pricing Power Inflation

Pricing power inflation is more often called as administered price inflation. This type of inflation occurs when the business houses and industries decide to increase the price of their respective goods and services to increase their profit margins.

Sectoral Inflation

This type of inflation takes place when there is an increase in the price of the goods and services produced by a certain sector of industries. For instance, an increase in the cost of crude oil would directly affect all the other sectors, which are directly related to the oil industry.

Effects of Inflation on Valuation of Stock

If prices were constant during the period, all three methods of valuation, namely FIFO, LIFO, and weighted average would produce the exact same result since each unit would have been purchased for an identical amount. But, since prices usually change, each method will produce different results. During periods of inflation, LIFO will generate a lower amount of gross profit and a lower inventory value. The FIFO method will produce a higher gross profit and a higher ending inventory balance. During periods of inflation, LIFO offers substantial tax savings due to lower profits and lower inventories. However, in periods of deflation, the effects are just the opposite.

Budget and Budgetary Control

Budgets are formal quantitative statements of the resources set aside for carrying out planned activities over given periods of time. As such, they are widely used means for planning and controlling activities at every level of the organization. There are a number of reasons for their wide range.

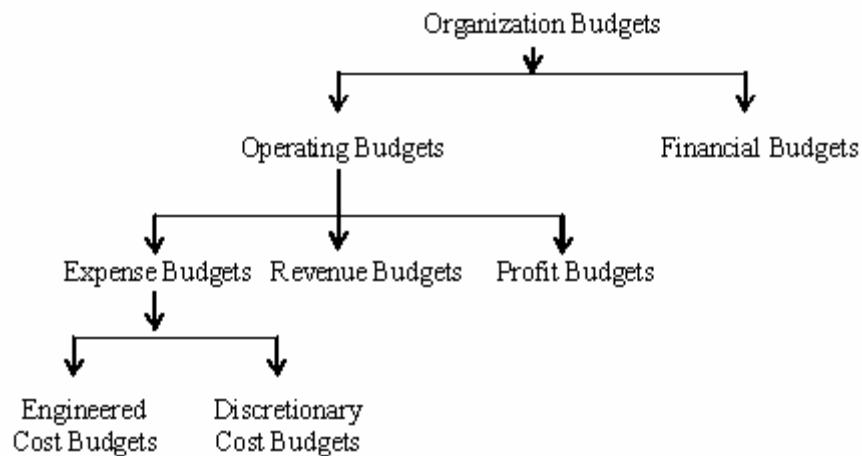
First, budgets are stated in monetary terms, which are easily used as a common denominator for a wide variety of organizational activities – hiring and training personnel, purchasing equipment, manufacturing, advertising and selling. Second, the monetary aspects of budgets means that they can directly convey information on a key

organizational resource – capital – and on a key organizational goal – profit. They are, therefore, heavily favored by profit-oriented companies.

Third, budgets establish clear and unambiguous standards of performance for a set time period – usually a year. At stated intervals during that time period, actual performance will be compared directly with the budget. Deviations can be detected quickly and acted upon.

In addition to being a major control device, budgets are one of the major means of co-ordinating the activities of the organization. The interaction between managers and subordinates that takes place during the budget development process will help define and integrate the activities of the organization members.

The different types of organization budgets are given below.



A brief description of the above classification is provided below.

- *Operating Budget*- Budget indicating the goods and services the organization expects to consume in a budget period
- *Financial Budget*- Budget detailing the money expected to be spent during the budget period and indicating its sources
- *Expense Budget*- Budget explaining where money was applied
- *Engineered Cost Budget*- Type of expense budget that describes material and labour cost of each item produced, including estimated overhead costs
- *Discretionary Cost Budget*- Type of expense budget that is used for departments in which output cannot be accurately measured

- *Revenue Budget*- Budget for projected sales revenue, used to measure marketing and sales effectiveness
- *Profit Budget or Master Budget*- Budget combining expense and revenue budgets in one unit

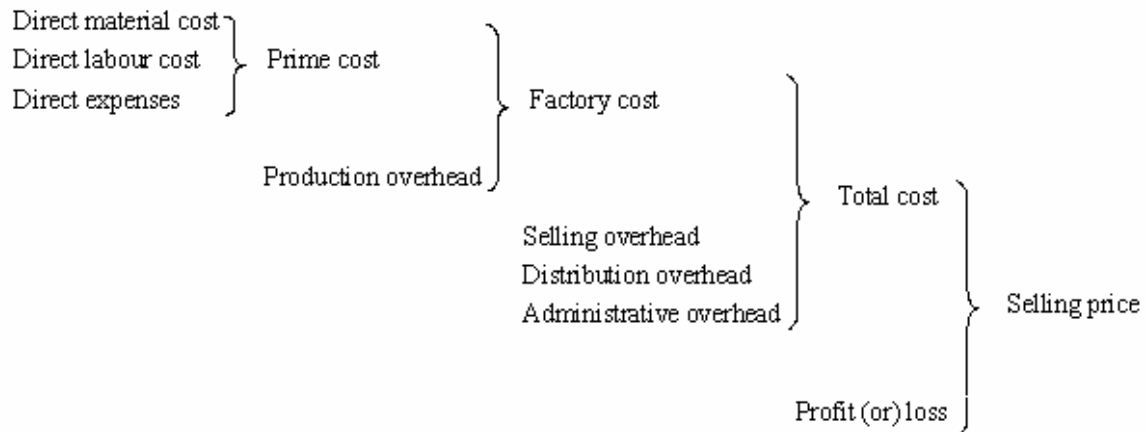
Elements of Cost

For the purpose of identification, accounting and control, breakup of cost into its elements is essential. Costs are normally broken down into three basic elements, namely, material, labour and expense. Material cost includes all materials consumed in the process of manufacture up to the primary packing. Labour cost includes all remuneration paid to the staff and workmen for conversion of raw materials into finished products. Expenses consist of the cost of utilities and services used for the conversion process including notional cost for the use of owned assets.

Each of the cost elements can be further divided into direct and indirect cost. Direct costs are those which can be identified with or related to the product or services, so much so that an increase or decrease of a unit of product or service will affect the cost proportionately. Indirect cost, on the other hand, cannot be identified or traced to a given cost object but are related to the expenses incurred for maintaining facilities for such production or services. Thus, the elements of cost may be summarised as follows – (a) Direct materials and indirect materials, (b) Direct wages and indirect wages, (c) Direct expense and indirect expense

The aggregate of direct materials cost, direct wages and direct expense is called *Prime cost*, while indirect materials cost, indirect wages and indirect expenses are collectively called *overhead cost*.

Overheads are classified into production overheads i.e. indirect costs relating to manufacturing activities, administration overheads i.e. costs relating to formulating the policy, directing the organisation and controlling operations and selling and distribution overheads i.e. indirect costs relating to the activity of creating and stimulating demand and securing orders as well as operations relating to distribution of goods from factory warehouse to customers. The following figure indicates the break up of various cost elements in order to arrive at the selling price.



Elements of cost

Allocation of Overheads

Manufacturing Costs or Operating Costs are equal to the direct production costs + fixed charges + plant overhead costs and the overhead cost refers to any costs not specifically or directly associated with the production of identifiable goods or services. Therefore in order to calculate the total cost of the product it is necessary to assign the overhead to the product.

If an industry is making only one product, a uniform charge for overhead may be possible. But if a number of different products are being manufactured, an equitable base must be sought out to charge each product with a fair and reasonable share of the overhead cost, so that the total cost of each unit may be calculated. The following are the different methods for the allocation of overheads :

1) Direct material cost method

$$\text{Overhead for an item} = \frac{\text{Factory overhead} \times \text{Direct material cost of the item concerned}}{\text{Total direct material cost}}$$

2) Direct labour cost method

$$\text{Overhead for an item} = \frac{\text{Factory overhead} \times \text{Direct labour cost of the item concerned}}{\text{Total direct labour cost}}$$

3) Prime cost method

$$\text{Overhead for an item} = \frac{\text{Factory overhead} \times \text{Prime cost of the item concerned}}{\text{Total prime cost}}$$

← Percentage overhead

Note: Prime cost = direct material cost + direct labour cost

4) Labour hour rate method

$$\text{Overhead for an item} = \frac{\text{Factory overhead} \times \text{Labour hours associated with the item concerned}}{\text{Total labour hours}}$$

↑
Rate per hour of direct labour

5) Machine hour rate method

$$\text{Overhead for an item} = \frac{\text{Factory overhead} \times \text{Machine hours associated with the item concerned}}{\text{Total machine hours}}$$

↑
Machine hour rate

6) Production Unit method

Overhead associated with a production order

$$\text{order} = \frac{\text{Factory overhead} \times \text{Number of units produced as per the production order}}{\text{Total production in terms of units}}$$

Problem

A company produces two items A and B and the details are as given below:

Item	No. produced / year	Material Cost / unit (₹)	Labour cost/ unit (₹)
A	20000	200	100
B	15000	350	150

The company's overhead per year is ₹ 5000000. Allocate the overhead and find the total cost per unit for A and B using the following methods.

- Proportional to material cost
- Proportional to labour cost
- Proportional to Material and Labour cost

Answer

$$\text{Overhead for item A based on material cost} = \frac{5000000 \times (200 \times 20000)}{200 \times 20000 + 350 \times 15000} = ₹ 2162162.1$$

$$\text{Overhead for item A per unit} = \frac{2162162.1}{20000} = ₹ 108.10$$

$$\begin{aligned} \text{Total cost of item A per unit} &= \text{Material cost} + \text{Labour cost} + \text{Overhead cost} \\ &= 200 + 100 + 108.1 = ₹ 408.1 \end{aligned}$$

Overhead for item B may be calculated as above. Alternately, it can be obtained by subtracting the overhead of item A from the total factory overhead.

$$\text{Overhead for item B based on material cost} = 5000000 - 2162162.1 = ₹ 2837837.9$$

$$\text{Overhead for item B per unit} = \frac{2837837.9}{15000} = ₹ 189.189$$

$$\begin{aligned} \text{Total cost of item B per unit} &= \text{Material cost} + \text{Labour cost} + \text{Overhead cost} \\ &= 350 + 150 + 189.189 = ₹ 689.189 \end{aligned}$$

Part (b) and Part (c) of the above question can be done on similar lines.

Problem

A fabrication and assembly shop had its total overheads of ₹ 10000. It used direct material worth ₹ 10000 and paid ₹ 15000 as direct labour charges. Calculate the percentage overhead. If one product has its prime cost as ₹ 5000 determine the overheads or on cost related to it.

Answer

$$\begin{aligned} \text{Percentage overhead} &= \frac{\text{Factory overhead}}{\text{Total prime cost}} \\ &= \frac{10000}{10000 + 15000} = 0.4 \text{ or } 40\% \end{aligned}$$

$$\text{Overhead for the product concerned} = \frac{10000 \times 5000}{25000} = ₹ 2000$$

Problem

A fitting and assembly shop had its factory overheads of ₹ 12000 and the production for the period in terms of direct labour was 24000 hours. If a particular job takes 20 labour hours, calculate the overhead applied.

Answer

$$\text{Overhead applied} = \frac{12000 \times 20}{24000} = ₹ 10$$

Problem

The estimated overhead cost of a factory making transistors is ₹ 8000 in a particular period and the number of transistors produced during this period is 400. If a particular production order requires 100 such transistors, determine the factory overhead to be applied to this production order. Also calculate the overhead rate per transistor.

Solution

$$\text{Overhead for the production order} = \frac{8000 \times 100}{400} = ₹ 2000$$

Number of transistors in the production order = 100

Therefore, overhead rate per transistor = $2000 / 100 = ₹20$

Standard Costing Vs. Marginal Costing

Standard cost is defined as a predetermined cost which is calculated from management standards of efficient operations and the relevant necessary expenditure. They are the predetermined costs on technical estimate of material labour and overhead for a selected period of time and for a prescribed set of working conditions. In other words, a standard cost is a planned cost for a unit of product or service rendered.

The technique of using standard costs for the purposes of cost control is known as standard costing. It is a system of cost accounting which is designed to find out how much should be the cost of a product under the existing conditions. The actual cost can be ascertained only when production is undertaken. The predetermined cost is compared to the actual cost and a variance between the two enables the management to take necessary corrective measures.

Marginal cost means the cost of the marginal or last unit produced. It is also defined as the cost of one more or one less unit produced besides existing level of production.

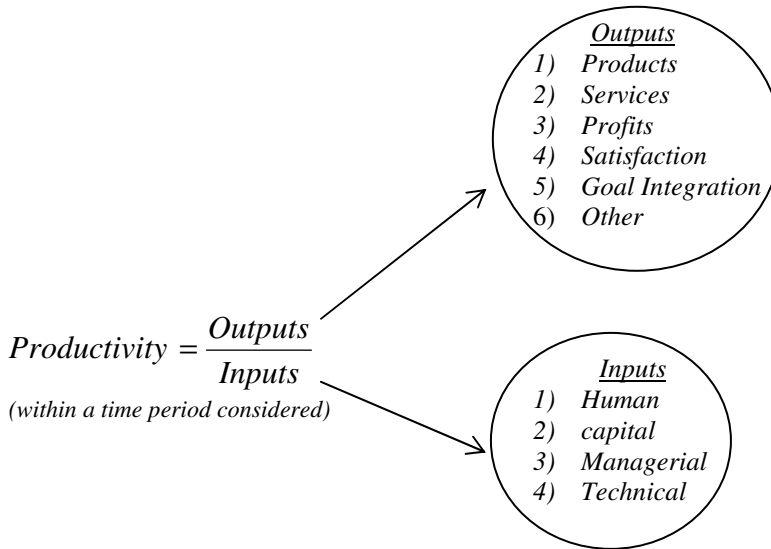
The marginal cost of a product is its variable cost. This is normally taken to be direct labour, direct material, direct expenses and the variable part of overheads. Marginal costing technique has given birth to a very useful concept of contribution where contribution is given by: Sales revenue less variable cost (marginal cost)

Contribution may be defined as the profit before the recovery of fixed costs. Thus, contribution goes toward the recovery of fixed cost and profit, and is equal to fixed cost plus profit ($C = F + P$). In case a firm neither makes profit nor suffers loss, contribution will be just equal to fixed cost ($C = F$). This is known as break even point.

Module IV

Productivity and Production

Productivity is the measure of how well the resources are brought together in an organization and utilized for accomplishing a given set of objectives. It can be defined as the output – input ratio within a time period with due consideration for quality. It can be expressed as follows:



The formula indicates that productivity can be improved by:

- 1) increasing the outputs with the same inputs
- 2) decreasing the inputs by maintaining the same outputs or
- 3) increasing the outputs and decreasing the inputs to change the ratio favourably.

Productivity implies effectiveness and efficiency in individual and organizational performance. Effectiveness is the achievement of objectives whereas efficiency is the achievement of ends with the least amount of resources. Managers cannot know whether they are productive unless they first know their objectives and goals and those of the organization.

Production and productivity are different terms and implies different meaning. It should be noted that higher production need not necessarily lead to higher productivity and vice versa.

Production is a process (or system) of converting input into some useful, value added output. Production is a measure of output produced. The emphasis is not on how well the input-resources are utilized. Productivity, on the other hand, puts emphasis on the ratio of output produced to the input used. That is, here the focus is on how well the input resource is used for conversion into output.

Problem

- A company is manufacturing 24,000 components per month by employing 100 workers in 8 hours shift. Calculate the productivity?
- Suppose the company gets additional order to supply 6000 more components and the management decides to employ additional workers, calculate the productivity when the number of additional workers employed are (i) 30 (ii) 25 and (iii) 20.

Answer

- a) Output in terms of production = 24000 components
 Input in terms of man-hours = $100 \text{ workers} \times 8 \text{ hours} \times 30 \text{ days}$
 (assuming 30 days in a month)
 = 24,000 man-hours

$$\begin{aligned}\text{Productivity} &= \frac{24,000 \text{ components}}{24,000 \text{ man-hours}} \\ &= 1 \frac{\text{component}}{\text{man-hour}}\end{aligned}$$

- b) (i) When additional 30 workers are employed

$$\begin{aligned}\text{Productivity} &= \frac{24,000 + 6,000}{(100 + 30)(8)(30)} \\ &= 0.96 \frac{\text{component}}{\text{man} - \text{hour}}\end{aligned}$$

- (ii) When additional 25 workers are employed

$$\begin{aligned}\text{Productivity} &= \frac{24,000 + 6,000}{(100 + 25)(8)(30)} \\ &= 1 \frac{\text{component}}{\text{man-hour}}\end{aligned}$$

- (iii) When additional 20 workers are employed

$$\begin{aligned}\text{Productivity} &= \frac{24,000 + 6,000}{(100 + 20)(8)(30)} \\ &= 1.04 \frac{\text{component}}{\text{man} - \text{hour}}\end{aligned}$$

Note:-

- 1) *Increase in production does not necessarily mean increase in productivity*
- 2) *Productivity is always associated with the context in which it is calculated. For example, we have calculated labour productivity*

Approaches to Measure Productivity

With respect to assessment methodology, one of the best-known measures of productivity, ***Partial Productivity Measure (PPM)***, relates total output to one class of input. Examples of industry-based partial productivity measures are listed in the following Table

Partial Productivity Measures

Partial Productivity	Mathematical Representation
Labour Productivity	Output / Labour Input
Material Productivity	Output / Material Input
Capital Productivity	Output / Capital Input
Energy Productivity	Output / Energy Input
Advertising and Media Planning Productivity	Output / Advertising and Media Planning Input
Other Expense Productivity	Output / Other Expense Input

A second, and relatively new, approach to the assessment of productivity, **Total Productivity Measure (TPM)**, relates total outputs to the sum of all tangible input factors (labour, materials, capital, energy, other expenses, etc.).

Total Productivity Measure (TPM) = Total outputs / Sum of all Inputs

A third measure of productivity, **Total Factor Productivity (TFP)**, relates total output to the sum of associated labor and capital inputs.

Total Factor Productivity (TFP) = Total Output / (Labour + Capital Inputs)

Note:-

- 1) The outputs of the firm as well as the inputs are usually expressed in a common measurement unit. It will be convenient to express them in terms of monetary value (Rupees)
- 2) To compare productivity, indices are adjusted to the base year and must be stated in terms of the base year Rupee value. This is referred to as deflating the input and output factors. Deflators are used to nullify the effect of changing price from one year to another year
- 3) Deflator = Current Year Price / Base Year Price
- 4) Example indicating the use of deflator in productivity calculation is given below:

Year 1	Year 2
Output in terms of Rupees = ₹ 20,000	Output in terms of Rupees = ₹ 20,000
Material Usage: 20 kg of material	Material Usage: 20 kg of material
Material Price: ₹ 50/kg.	Material Price: ₹ 70/kg.
Material Productivity: $\frac{20,000}{20 \times 50} = 20$	Material Productivity: $\frac{20,000}{20 \times 70} = 14.28$ (if deflator is not considered and is not realistic) Deflator = $\frac{70}{50} = 1.4$ Considering the deflator, Material input = $\frac{20 \times 70}{1.4} = ₹ 1000$. and Material Productivity: $\frac{20,000}{1000} = 20$
That is, for the same output produced with the same amount of input the productivity should not vary with time	

Problem

Following information pertains to a firm's performance during the last four time periods. Compute the partial productivity and total productivity ratios and also productivity indices for each of the four time periods. Assume period 1 as the base year.

Particulars	Period 1	Period 2	Period 3	Period 4
Outputs				
1. Finished goods	2500	2200	2800	3200
Work in process	1200	1600	1000	4000
% Completion	60%	50%	30%	15%
Price per unit (₹)	1000	1200	1500	1700
2. Dividend from securities (₹.)	12,000	15,000	28,000	29,000
Deflator for item (2)	1	1.11	1.12	1.5
Inputs				
1. Skilled labour (hours)	10,000	12,000	12,000	10,000
Average wage rate (₹)	60	70	75	75
2. Unskilled labour (hours)	5000	8000	5000	3000
Wage rate (₹)	30	40	40	50
3. Raw materials (tones)	20	18	23	25
Price per tonne (₹)	1200	1600	2000	2000
4. Total plant hours worked	1800	2400	2500	2500
Plant hour rate (₹)	650	650	1500	2400
5. Energy				
a) Oil used (litres)	5000	3000	2000	1500
Price per litre (₹)	4	6	8	12
b) Coal (tones)	200	150	50	-
Price per tonne (₹)	1200	1800	2800	-
c) Electricity (kWH)	15,000	18,000	22,000	30,000
Rate per kWH (₹)	2.5	3.2	4.2	6.7
6. Other Expenses				
a) Consulting fees (₹)	20,000	-	40,000	2,00,000
b) Information Expenses (₹)	10,000	15,000	28,000	50,000
Deflator for item (6)	1	1.2	1.3	1.3

Answer*Sample calculation (Period 1)*

$$\text{Output} = (2500 \times 1000) + (1200(0.6) \times 1000) + 12000 = ₹ 3232000$$

$$\text{Labour Input} = (10,000 \times 60) + (5000 \times 30) = ₹ 750000$$

$$\text{Labour Productivity Ratio (LPR)} = 4.3$$

Sample calculation (Period 2)

$$\text{Output} = \frac{2200 \times 1200}{1.2} + \frac{(1600(0.5) \times 1200)}{1.2} + \frac{15000}{1.11} = ₹ 3013513.50$$

$$\text{Labour Input} = \frac{(12,000 \times 70)}{1.16} + \frac{8000 \times 40}{1.33} = ₹ 964739.43$$

$$\text{Labour Productivity Ratio (LPR)} = 3.12$$

Capital Input → Plant hour cost

Continue.....

Productivity Ratios

Particulars	Period 1	Period 2	Period 3	Period 4
Total Productivity Ratio (TPR)	1.36	1.04	1.1	1.37
Labour Productivity Ratio (LPR)	4.3	3.12	3.59	5.53
Material Productivity Ratio (MPR)	134.6	139.2	112.8	126.8
Capital Productivity Ratio (CPR)	2.76	1.93	1.91	2.34
Energy Productivity Ratio (EPR)	10.86	12.71	25.22	47.15
Other Expenses Productivity Ratio	107.73	241.8	59.74	19.86

Productivity Indices

Particulars	Period 1	Period 2	Period 3	Period 4
Total Productivity Index (TPI)	1	0.76	0.81	1.01
Labour Productivity Index (LPI)	1	0.73	0.83	1.02
Material Productivity Index (MPI)	1	1.04	0.84	0.94
Capital Productivity Index (CPI)	1	0.7	0.69	0.85
Energy Productivity Index (EPI)	1	1.17	2.32	4.34
Other Expenses Productivity Index	1	2.23	0.56	0.19

Advantages and Disadvantages of Different Productivity Measures

Partial Productivity Measure

Advantages	Disadvantages
1) <i>Easy to understand and calculate</i> 2) <i>A tool to pinpoint improvement</i>	1) <i>Misleading if used alone</i> 2) <i>No consideration for overall impact</i>

Total Productivity Measure

Advantages	Disadvantages
1) <i>More accurate representation of the total picture of the company</i> 2) <i>Easily related to total costs</i> 3) <i>Considers all quantifiable outputs and inputs</i>	1) <i>Difficulty in obtaining the data</i> 2) <i>Requirement of special data collection system</i>

Total Factor Productivity

Advantages	Disadvantages
1) <i>Data required for the calculation is relatively easy to obtain from company records</i> 2) <i>Value added approach</i>	1) <i>No consideration for material and energy input</i> 2) <i>Difficult to relate value added approach to production efficiency</i>

Factors Influencing Productivity

Controllable factors (Internal Factors) 1) <i>Product</i> 2) <i>Plant and equipment</i> 3) <i>Technology</i> 4) <i>Materials</i> 5) <i>Human factors</i> 6) <i>Work methods</i> 7) <i>Management style</i> 8) <i>Financial factors</i> Uncontrollable factors (External factors) 1) <i>Natural Resources</i> 2) <i>Government Policy</i> 3) <i>Political Stability</i>

Ways to Improve Productivity

Productivity of any system can be improved by proper use of resources and optimum utilization of system or processes. These include the following.

Technology Based

- a) *Acquire new technology*
- b) *Automation in assembly*
- c) *Modern maintenance techniques*
- d) *Energy technology*
- e) *Flexible Manufacturing Systems (FMS)*

Employee Based

- a) *Financial and non-financial incentives at individual and group level*
- b) *Suggestion scheme*
- c) *Safe work-place*
- d) *Workers participation in management*

Material Based

- a) *Material Planning and control*
- b) *Purchasing, logistics*
- c) *Material storage and retrieval*
- d) *Source selection and procurement of quality material*
- e) *Waste elimination*
- f) *Use of Automated Guided Vehicle (AGV) for material transportation*

Process Based

- a) *Methods engineering and work simplification*
- b) *Job design, Job evaluation, Job safety*
- c) *Human factors engineering*

Product Based

- a) *Value analysis and value engineering*
- b) *Product diversification*
- c) *Standardization and simplification*
- d) *Reliability engineering*
- e) *Promotion*

Management Based

- a) *Management style*
- b) *Communication in the organization*
- c) *Work culture*
- d) *Motivation*
- e) *Promoting group activity*

Material Management

Materials management is concerned with planning, directing and controlling the kind, amount, location, movement and timing of various flow of materials used in and produced by the process.

Objectives of Materials Management

- 1) Making available supply of materials in specified quantity and quality at economic cost
- 2) Maintaining the continuity of supply
- 3) Optimization of investments in materials and inventory cost
- 4) Assuring high inventory turnover
- 5) Purchasing the items from a reliable source at economic price
- 6) Reduction of costs by using various cost reduction techniques such as variety reduction, standardization and simplification, value analysis, inventory control, purchase research, etc.
- 7) Co-ordination of the functions such as planning, scheduling, storage and maintenance of materials

Scope of Materials Management

- 1) Materials planning and control
- 2) Store-keeping
- 3) Inventory control
- 4) Simplification, codification, and standardization in stores
- 5) Transportation
- 6) Material handling
- 7) Disposal of scrap and surplus

Inventory Control (Material Control)

Inventory is money kept in the store room in the form of raw material, in-process material and finished goods.

Most of the production activities are engaged in modifying the material to create an end product. Since material is the fundamental component of most activities it is extremely necessary for the production control system to provide planning and control of material so that right material (a good quality one) is available in right quantity at the right time.

There are three major classes of inventory, namely

- 1) Raw materials and purchased spare parts.
- 2) In-process material which means semi finished goods.
- 3) Finished goods which are lying in stock room waiting for dispatch.

Although inventory is an idle resource (money blocking) it is a must for smooth running of the organization.

Inventory management basically deals with the following two problems:

- 1) The amount of material to be ordered (order quantity)
- 2) When the material is to be ordered (ordering point)

Economic Order Quantity (EOQ)

The economic order quantity is the size of an inventory order which minimizes the inventory cost. The inventory cost is the sum of procurement cost and carrying cost.

To determine EOQ two extreme views are encountered:

- 1) Order for very large lots (Produce in very large lots) to minimize the procurement cost (to minimize set up cost)
- 2) Order for every small lots (produce in very small lots) to minimize the storage cost or carrying cost.

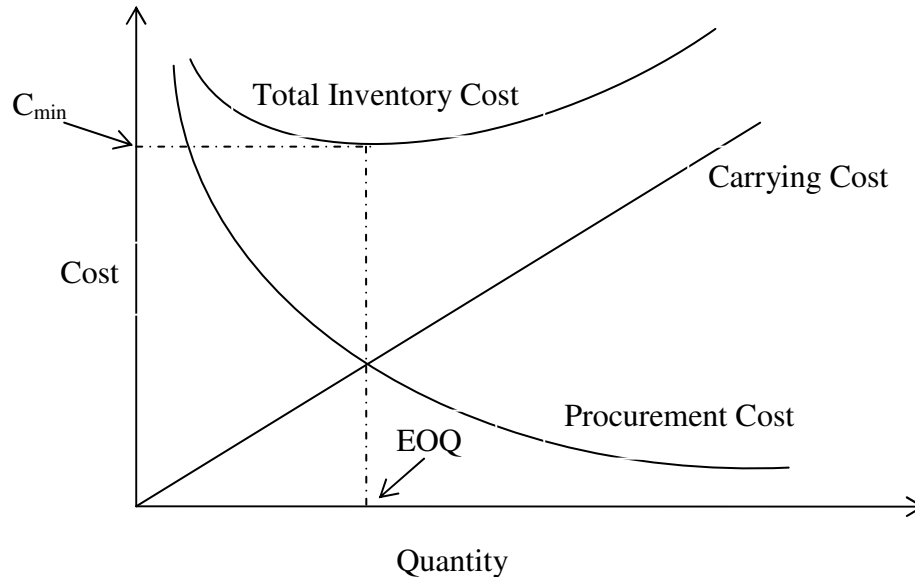


Figure indicating the variation of procurement cost and carrying cost with order quantity

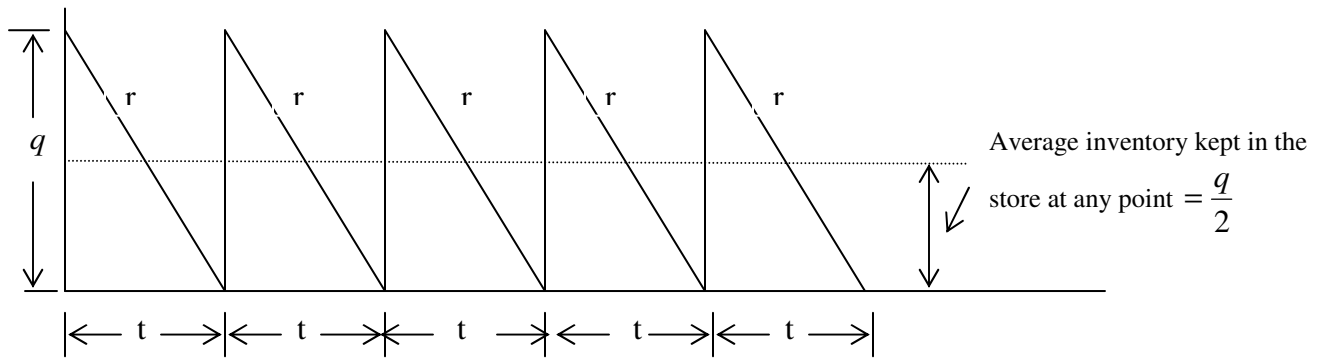
The variation of procurement cost and carrying cost with order quantity is shown in the above figure and from the graph it can be seen that the interaction of these cost elements will lead to an optimum point. The quantity corresponding to this point is called the Economic Order Quantity (EOQ) and corresponding cost will be the minimum cost of inventory. The condition for economical order quantity (EOQ), which minimizes the total inventory cost is:

$$\text{Procurement cost} = \text{Carrying cost}$$

Purchasing Model with No Shortage

Assumptions

- 1) Fixed demand rate (fixed production rate)
- 2) Instantaneous replacement (Lead time = 0)
- 3) No shortage is permitted.



Let, C_3 = Procurement cost / order (Setup cost / order)

r = Demand rate in units / year

C_1 = Carrying cost / unit / year (C_1 is generally expressed as a fraction of cost of unit)

q = Quantity of units ordered at time t

(Note: One year is considered as the unit time)

No. of orders placed in a year = $\frac{r}{q}$

Therefore, Procurement cost / year = $\left(\frac{r}{q}\right)C_3$

Total carrying cost / year = $\left(\frac{q}{2}\right)C_1$

Therefore, Inventory cost/ year, C = procurement cost / year + carrying cost / year

$$C = \left(\frac{r}{q}\right)C_3 + \left(\frac{q}{2}\right)C_1$$

To minimize 'C', $\frac{dC}{dq} = 0$

$$\Rightarrow -\frac{rC_3}{q^2} + \frac{C_1}{2} = 0$$

$$\Rightarrow q = \sqrt{\frac{2C_3r}{C_1}} = EOQ \text{ (Economical Order Quantity)}$$

Substituting the value of EOQ in the cost equation, $C = \left(\frac{r}{q}\right)C_3 + \left(\frac{q}{2}\right)C_1$, the minimum

inventory cost can be expressed as $C_{\min} = \sqrt{2C_1C_3r}$

$$\text{Economic Order Quantity, } EOQ = \sqrt{\frac{2C_3r}{C_1}}$$

$$\text{Minimum Cost of Inventory, } C_{\min} = \sqrt{2C_1C_3r}$$

Problem

A manufacturer has to supply his customers with 600 units of his products per year. Shortages are not allowed and the storage cost amounts to ₹ 0.60 per unit per year. The setup cost per run ₹ 80. Find the optimum run size, its number and the minimum yearly cost of inventory.

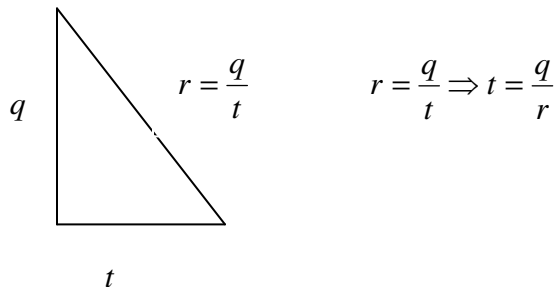
Answer

$$C_1 = ₹ 0.60 / \text{year}$$

$$C_3 = ₹ 80$$

$$r = 600 \text{ units / year}$$

$$\text{Optimum run size, } EOQ = \sqrt{\frac{2C_3r}{C_1}} = 400 \text{ units}$$



$$t = \frac{q}{r} = \frac{400}{600} = \frac{2}{3} \text{ years} = \frac{2}{3} \times 12 = 8 \cdot \text{months}$$

$$\text{Minimum Cost of Inventory, } C_{\min} = \sqrt{2C_1C_3r} = ₹ 240 / \text{year}$$

That is, the manufacturer has to produce **400 units** of his products once in **8 months** and the minimum inventory cost works out to be **₹ 240 / year**.

Problem

The storage cost of an item is ₹1 per month and the setup cost is ₹25 per run. If the production is instantaneous and the demand is 200 units per month find the optimum size of the batch and the best time for replenishment of inventory.

Answer

$$C_1 = ₹1 / \text{month}$$

$$C_3 = ₹ 25$$

$$r = 200 \text{ units / month}$$

$$\text{Optimum batch size, } EOQ = \sqrt{\frac{2C_3r}{C_1}} = 100 \text{ units}$$

$$t = \frac{q}{r} = \frac{100}{200} = \frac{1}{2} \text{ month} = \frac{1}{2} \times 30 = 15 \cdot \text{days}$$

$$\text{Minimum Cost of Inventory, } C_{\min} = \sqrt{2C_1C_3r} = ₹100 / \text{month}$$

That is, the industry should replenish the inventory once in **15 days** and the minimum inventory cost works out to be **₹ 100 per month**. The optimum size of the batch is **100 units**.

Problem

You have to supply your customer with 100 units of a product every Monday. You obtain the product from a local supplier at ₹60 per unit. The cost of ordering and transporting is ₹150 per order. The cost of carrying inventory is estimated at 15 % per year of the cost of the product carried. Find the lot size that will minimize the cost of the system and determine the total cost that you incur per week.

Answer

$$C_1 = 15 \% \text{ of the cost of the product / year}$$

$$= ₹ 0.15 \times 60 / \text{unit/year}$$

$$= ₹ \frac{0.15 \times 60}{52} / \text{unit / week}$$

$$C_3 = \text{Rs. } 150 / \text{order}$$

$$\text{Economic Order Quantity, } EOQ = \sqrt{\frac{2C_3r}{C_1}} = 416 \cdot \text{units}$$

$$\text{Minimum Cost of Inventory, } C_{\min} = \sqrt{2C_1C_3r} = ₹72 / \text{week}$$

$$\begin{aligned} \text{Total cost} &= \text{inventory cost} + \text{demand rate} \times \text{cost of the product / unit} \\ &= 72 + 100 \times 60 = ₹ 6072 / \text{week} \end{aligned}$$

Problem

Calculate EOQ from the following data:

Annual demand = 1600 units

Cost of materials / unit = ₹ 40

Cost of placing and receiving an order = ₹ 50

Annual carrying cost of inventory = 10 % of inventory value

Answer

$$C_1 = 40 \times 0.1 = ₹ 4 / \text{unit} / \text{year}$$

$$C_3 = ₹ 50$$

$$r = 1600 \text{ units} / \text{year}$$

$$\text{Economic Order Quantity, } EOQ = \sqrt{\frac{2C_3r}{C_1}}$$

Problem

ABC manufacturing company purchases 9000 parts of a machine for its annual requirement ordering one month usage at a time. Each part costs ₹20/-. The ordering cost per order is ₹15/- and the carrying charges are 15% of the average inventory cost per year. Suggest a more economical purchasing policy for the company and how much the company saves per year with your advice. Derive the formula used, if any.

Answer

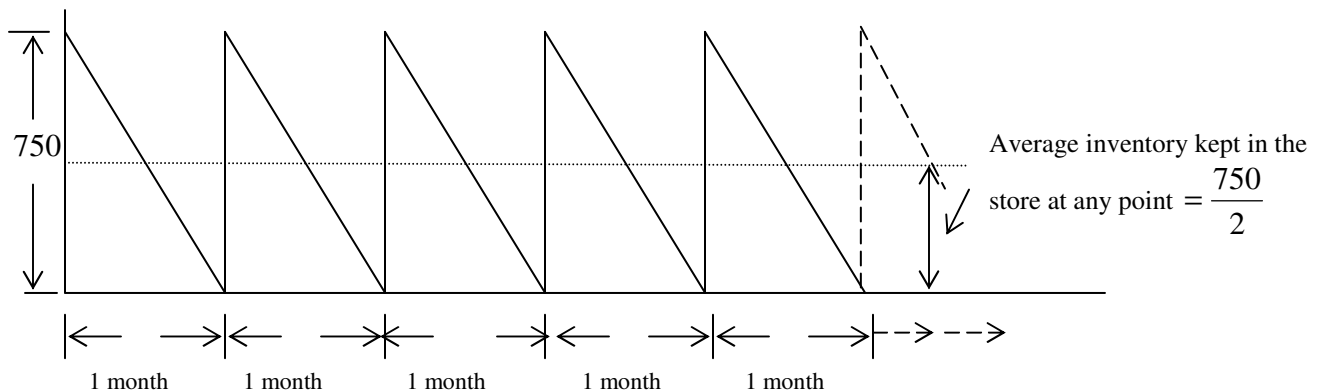
Inventory cost per year going by the policy of ordering once in a month

$$\text{Cost / order} = ₹ 15$$

$$\text{Total No. of orders} = 12$$

$$\text{Total ordering cost (or procurement cost)} = 12 \times 15 = ₹ 180 \text{ (per year)}$$

$$\text{Lot size per month} = \frac{9000}{12} = 750 \text{ units}$$



$$\text{Carrying cost / year} = 375(0.15 \times 20) = ₹ 1125$$

$$\text{Therefore, total inventory cost per year} = 180 + 1125 = ₹ 1305 / \text{year}$$

Adopting the basic inventory model

$$\text{Economic Order Quantity, } EOQ = \sqrt{\frac{2C_3r}{C_1}} = \sqrt{\frac{2 \times 15 \times 9000}{0.15 \times 20}} = 300 \text{ units}$$

$$\text{Ordering time, } t = \frac{q}{r} = \frac{300}{9000} = \frac{1}{30} \text{ year} \approx 12 \text{ days}$$

$$\text{Cost of Inventory, } C_{\min} = \sqrt{2(0.15 \times 20)15 \times 9000} = ₹900 / \text{year}$$

$$\text{Savings} = 1305 - 900 = ₹ 405 \text{ per year}$$

Hence the company should order 300 parts at time interval of roughly 12 days in place of ordering 750 units per month, which results in a yearly saving of ₹ 405.

Stores Management

Stores management takes care

1. that the required material is never out of stock;
2. to purchase materials on the principle of economic order quantity, so that the associated costs can be minimized.
3. to protect stores against damage, theft, etc.

Types of stores

Stores can be of two types, namely decentralized and centralized Stores. In decentralized stores system, each section of the industry (e.g. foundry, machine shop, forging, etc.) has separate store attached with it, whereas in centralized stores system, the main store located centrally fulfills the needs for each and every department.

Advantages of centralization of Stores

- 1) *Better supervision and control*
- 2) *It requires fewer personnel to manage and thus involves reduced related costs.*
- 3) *Better layout of stores*
- 4) *Minimum stores can be maintained*
- 5) *Better security arrangements can be made*

Advantages of Decentralization of Stores

- 1) *Reduced material handling and the associated cost.*
- 2) *Convenient for very department to draw materials,.*
- 3) *Less chances of production stoppages owing to easy and prompt availability of materials.*

Examples for both types of stores are given in the following figures

.

Moulding & Casting	Forging	Heat Treatment	Machine Shop
Foundry	STORE	Packing	
Pattern Making			
Fitting Shop	Office		

Centralized Store Keeping

Moulding & Casting		S*	Forging	S*
		S*		Machine Shop
Pattern Making		Office		S*
S*	S*			Packing
Fitting Shop				

S* - Store

Decentralized Store Keeping

Bin Cards

A bin is an open-mouth container made of steel or wood. A bin card is a card attached to each bin or rack or shelf, used for storing materials and the like. The Bin Card shows details of quantities of each types of material received, issued and on hand each day. The store-keeper maintains bin cards up to date. A bin card is not considered as an accounting record; it simply informs store-keeper of the quantities of each item on hand. Following figure shows a BIN (or STOCK) CARD.

Lecture Notes on Industrial Organization and Management

*Subject handled by: Dr. Shouri P.V., Associate Professor in Mechanical Engineering, MEC, Cochin
(for S₇ Computer Science Batches)*

BIN CARD				
Bin No.		Maximum Quantity		
Material		Ordering Level		
Code No.....		Minimum Quantity		
Stores Ledger Folio				
Date	Quantity received	Quantity issued	Balance	Remarks

Bin cards may be in duplicates. One card is attached to the bin containing materials. The duplicate card remains with the store-keeper, on his table for ready reference. Bin card helps the store-keeper to know about the details of the stock of materials. When the quantity reaches a minimum value, he can place orders or requisitions for fresh supply of materials.

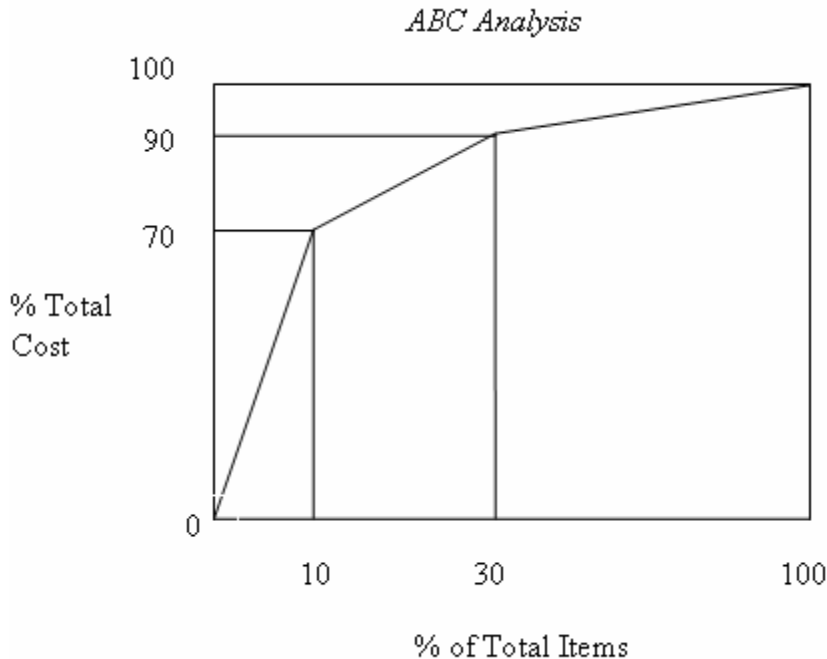
Bin card may also have the following details:

- 1) The maximum and minimum quantity of each material to be carried out.
- 2) Normal quantity of each material to be ordered.
- 3) “Ordering Level” of the material.

Note: Stores Ledger is identical with Bin Card except that money values are shown

ABC Analysis

ABC analysis, also known as Selective Inventory Control is a business term used to define an inventory categorization technique often used in materials management. It can be observed that in a typical industrial situation nearly 10% of items have 70% of the annual inventory consumption, 20% of the items have 20% of annual inventory consumption, and 70% of the items have only 10% of the annual inventory consumption. Since 70% of the annual consumption of inventory is covered by only 10% of the items in the inventory, these items deserve highest attention and are classified as ‘A’ items. Similarly 20% of the items covering 20 % of the inventory investment are B class items and require moderate attention. Balance 70% of the inventory items are termed as C class items and require least attention as they consume only 10% of the total inventory cost.



Steps in ABC analysis

The steps in computing ABC analysis are:

- 1) Determine the annual usage in units for each item for the past one-year.
- 2) Multiply the annual usage quantity with the average unit price of each item to calculate the annual usage for each item in monetary terms.
- 3) Item with highest money usage annually is ranked first. Then the next lower annual usage item is listed till the lowest item is listed in the last.
- 4) Arrange the items in the inventory by cumulative annual usage (Rupees) and by cumulative percentage. Categorize the items in A, B, and C categories.

Advantages of ABC analysis

- 1) A Close and strict control is facilitated on the most important items which constitute a major portion of overall inventory valuation or overall material consumption & due to this, costs associated with inventories maybe reduced.
- 2) The investment in inventory can be regulated in proper manner & optimum utilization of available funds can be assured.
- 3) A strict control on inventory items in this manner helps in maintaining a high inventory turnover rates.

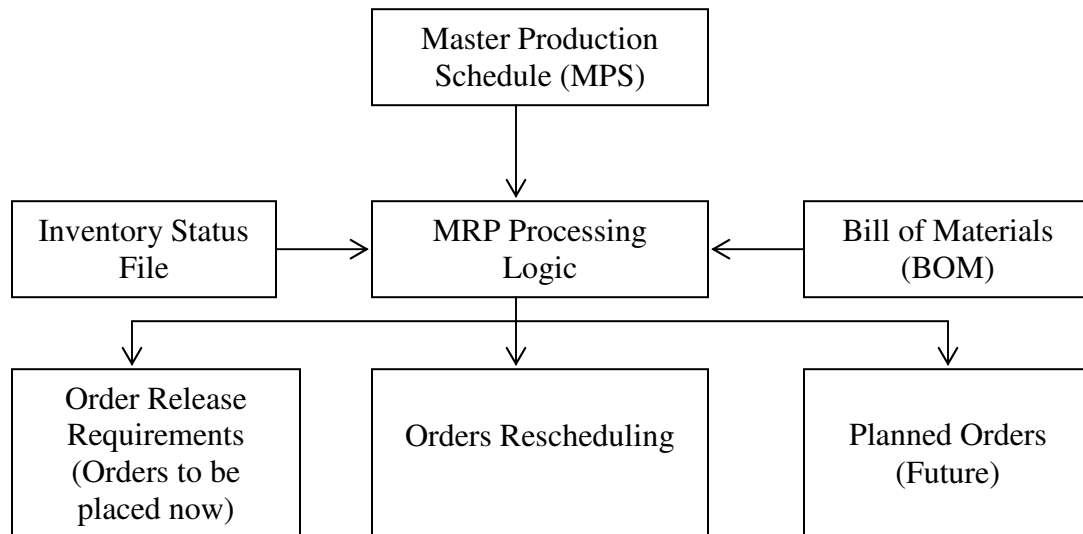
Material Requirement Planning

Material Requirement Planning (MRP) refers to the basic calculations used to determine component requirements from end item requirements. It also refers to a broader information system that uses the dependence relation ship to plan and control manufacturing operations. MRP is a technique for determining the quantity and timing for the acquisition of dependent demand items needed to satisfy master production schedule requirements.

MRP Objectives

- 1) Inventory reduction
- 2) Reduction in the manufacturing and delivery lead times
- 3) Realistic delivery commitments
- 4) Increased efficiency

MRP System



The above figure shows the MRP system. The inputs to the MRP system are:

- 1) A master schedule
- 2) An inventory status file, and
- 3) Bill of materials

Using these three information sources, the MRP processing logic (computer program) provides three kinds of information (output) for each product component:

- 1) Order release requirements
- 2) Order rescheduling, and
- 3) Planned orders

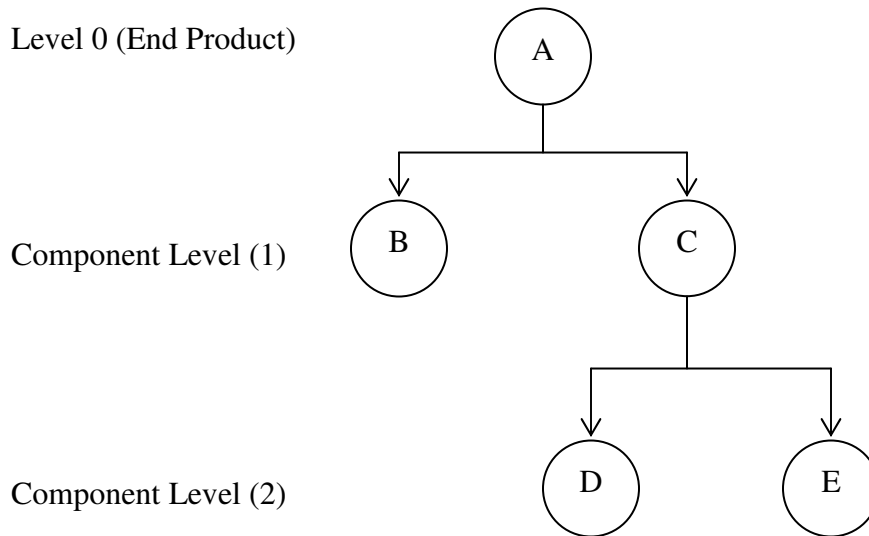
Master Production Schedule (MPS) is a series of time phased quantities for each item that a company produces, indicating how many are to be produced and when. MPS is initially developed from firm customer orders or from forecasts of demand before MRP system begins to operate. The MRP system accepts whatever the master schedule demands and translates MPS end items into specific component requirements. Most systems then make a simulated trial run to determine whether the proposed master schedule can be satisfied. Following figure shows a typical master schedule.

Master Production Schedule

Week	August 2010			
	1	2	3	4
Product P ₆		70		80
Product P ₈	40		75	60

Every inventory item being planned must have an **inventory status file** which gives complete and up to date information about the on hand quantities, gross requirements, scheduled receipts and planned order releases for the item. It also includes planning information such as lot sizes, lead times, safety stock levels and scrap allowances

To schedule the production of an end product, the MRP system must plan for all the materials, parts and subassemblies that go into the end product. The **Bill of Materials (BOM)** file in the computer provides this information. BOM file identifies each component by unique part number and help explode end item requirements into component requirements. The primary information to MRP from BOM is the product structure, the level of components to produce an end product. Following figure shows the product structure for the end item A, which is made up of component B and C. Further component C is made up of component D and E.

*Product structure for product A*

Benefits of MRP

- 1) Improved customer service
- 2) Reduction in lead time
- 3) Reduction in work- in- process
- 4) Reduction in past due orders
- 5) Elimination of annual inventory

- 6) Reduction in safety stock
- 7) Increase in productivity
- 8) Capacity constraints are better understood
- 9) Inventory turn-over increases

Problem

Complete the material requirement plan for an item X shown below. The item has an independent demand and a safety stock of 40 is maintained.

Order quantity = 70 Lead time = 4 Weeks Safety stock = 40	Week											
	1	2	3	4	5	6	7	8	9	10	11	12
Projected requirement	20	20	25	20	20	25	20	20	30	25	25	25
Receipts		70										
On hand at the end of the period (65)												
Planned order release												

Answer

Order quantity = 70 Lead time = 4 Weeks Safety stock = 40	Week											
	1	2	3	4	5	6	7	8	9	10	11	12
Projected requirement	20	20	25	20	20	25	20	20	30	25	25	25
Receipts		70			70			70			70	
On hand at the end of the period (65)	45	95	70	50	⁽³⁰⁾ 100	75	55	⁽³⁵⁾ 105	75	50	⁽²⁵⁾ 95	70
Planned order release	70			70			70					

Problem

A small scale industry manufactures a product and is expected to supply 80 units in week 1, 120 in week 4, 120 in week 6, and 100 in week 8. Each product is made up of 2 bearings, a shaft assembly, and one wheel. For these components order quantities, lead times and inventories on hand at the beginning of the period 1 are given below.

Part	Order Quantity	Lead Time	Inventory on hand
Bearing	600	2 weeks	200
Shaft Assembly	400	3 weeks	440
Wheel	800	1 week	100

Apart from the above requirement, another 180 shaft assembly is required for another customer in week 5, and 600 units of bearings are already scheduled to be received at the beginning of the week 2. Complete the material requirement plan for bearing, shaft, and wheel and show what quantities of orders must be released and when they must be released in order to satisfy MPS.

Answer

End Item Master Schedule

Week No.	1	2	3	4	5	6	7	8
Requirement	80			120		120		100

MRP – Bearing

Order quantity = 600 Lead time = 2Weeks	Week							
	1	2	3	4	5	6	7	8
Projected requirement	160			240		240		200
Receipts		600*						600
On hand at the end of the period (200)	40	640	640	400	400	160	160	(-40) 560
Planned order release						600		

MRP – Shaft Assembly

Order quantity = 400 Lead time = 3Weeks	Week							
	1	2	3	4	5	6	7	8
Projected requirement	80			120	180*	120		100
Receipts						400		
On hand at the end of the period (440)	360	360	360	240	60	(-60) 340	340	240
Planned order release			400					

MRP – Wheel

Order quantity = 800 Lead time = 1Week	Week							
	1	2	3	4	5	6	7	8
Projected requirement	80			120		120		100
Receipts				800				
On hand at the end of the period (100)	20	20	20	(-100) 700	700	580	580	480
Planned order release			800					